

A statement of type (iii) will appear in the definition of limit; however, it is necessary to change our point of view to some extent. To arrive at (iii), we first considered the domain of f and assigned values to x that were close to 4. We then noted the closeness of $f(x)$ to 5.5. In the definition of limit we shall *reverse* this process by first considering an open interval $(5.5 - \varepsilon, 5.5 + \varepsilon)$ and then, second, determining whether there is an open interval of the form $(4 - \delta, 4 + \delta)$ in the domain of f such that (iii) is true.

The next definition is patterned after the previous remarks.

Definition of Limit (2.3)

Let f be a function that is defined on an open interval containing a , except possibly at a itself, and let L be a real number. The statement

$$\lim_{x \rightarrow a} f(x) = L$$

means that for every $\varepsilon > 0$, there exists a $\delta > 0$, such that

$$\text{if } 0 < |x - a| < \delta, \text{ then } |f(x) - L| < \varepsilon.$$

If $\lim_{x \rightarrow a} f(x) = L$, we say that **the limit of $f(x)$, as x approaches a , is L** . Since ε can be arbitrarily small, the last two inequalities in this definition are sometimes phrased *$f(x)$ can be made arbitrarily close to L by choosing x sufficiently close to a* .

It is sometimes convenient to use the following form of Definition (2.3), where the two inequalities involving absolute values have been stated in terms of open intervals.

Alternative Definition of Limit (2.4)

The statement

$$\lim_{x \rightarrow a} f(x) = L$$

means that for every $\varepsilon > 0$, there exists a $\delta > 0$, such that if x is in the open interval $(a - \delta, a + \delta)$, and $x \neq a$, then $f(x)$ is in the open interval $(L - \varepsilon, L + \varepsilon)$.

To get a better understanding of the relationship between the positive numbers ε and δ in Definitions (2.3) and (2.4), let us consider a geometric interpretation similar to that in Figure 1.26, where the domain of f is represented by certain points on a coordinate line l , and the range by other points on a coordinate line l' . The limit process may be outlined as follows.

To prove that $\lim_{x \rightarrow a} f(x) = L$.

Step 1. For any $\varepsilon > 0$ consider the open interval $(L - \varepsilon, L + \varepsilon)$ (see Figure 2.5).



FIGURE 2.5

Step 2. Show that there exists an open interval $(a - \delta, a + \delta)$ in the domain of f such that (2.4) is true (see Figure 2.6).

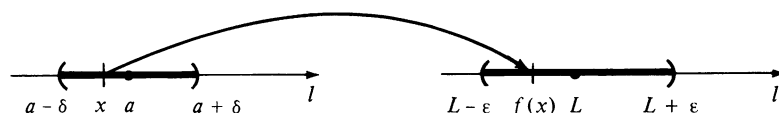


FIGURE 2.6

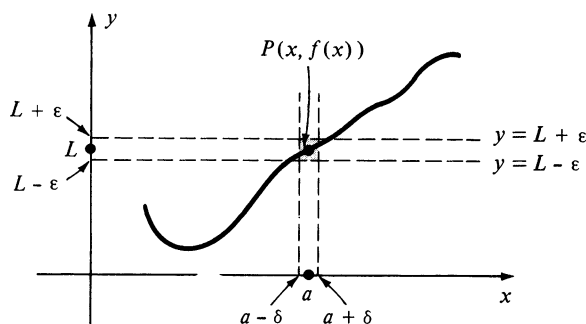
It is extremely important to remember that *first* we consider the interval $(L - \varepsilon, L + \varepsilon)$ and then, *second*, we show that an interval $(a - \delta, a + \delta)$ of the required type exists in the domain of f . One scheme for remembering the proper sequence of events is to think of the function f as a cannon that shoots a cannonball from the point on l with coordinate x to the point on l' with coordinate $f(x)$, as illustrated by the curved arrow in Figure 2.6. Step 1 may then be regarded as setting up a target of radius ε with bull's eye at L . To apply Step 2 we must find an open interval containing a in which to place the cannon such that the cannonball hits the target. Incidentally, there is no guarantee that it will hit the bull's eye; however, if $\lim_{x \rightarrow a} f(x) = L$ we can make the cannonball land as close as we please to the bull's eye.

It should be clear that the number δ in the limit definition is not unique, for if a specific δ can be found, then any *smaller* positive number δ' will also satisfy the requirements.

Since a function may be described geometrically by means of a graph on a rectangular coordinate system, it is of interest to interpret Definitions (2.3) and (2.4) graphically. Figure 2.7 illustrates the graph of a function f where, for any x in the domain of f , the number $f(x)$ is the y -coordinate of the point on the graph with x -coordinate x . Given any $\varepsilon > 0$, we consider the open interval $(L - \varepsilon, L + \varepsilon)$ on the y -axis, and the horizontal lines $y = L \pm \varepsilon$ shown in the figure. If there exists an open interval $(a - \delta, a + \delta)$ such that for every x in $(a - \delta, a + \delta)$, with the possible exception of $x = a$, the point $P(x, f(x))$ lies between the horizontal lines, that is, within the shaded rectangle shown in Figure 2.7, then

$$L - \varepsilon < f(x) < L + \varepsilon$$

and hence $\lim_{x \rightarrow a} f(x) = L$.

FIGURE 2.7 $\lim_{x \rightarrow a} f(x) = L$

The next example illustrates how the geometric process pictured in Figure 2.7 may be applied to a specific function.

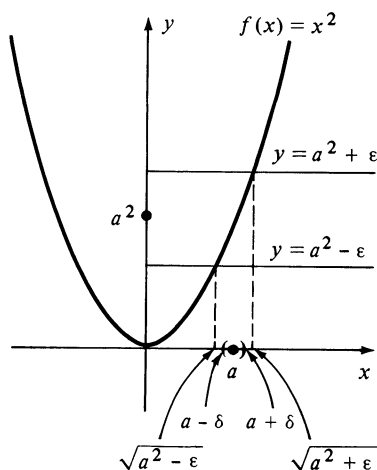


FIGURE 2.8

Example 1 Prove that $\lim_{x \rightarrow a} x^2 = a^2$.

Solution Let us consider the case $a > 0$. We shall apply (2.4) with $f(x) = x^2$ and $L = a^2$. The graph of f is sketched in Figure 2.8, together with typical points on the x - and y -axes corresponding to a and a^2 , respectively.

For any positive number ϵ , consider the horizontal lines $y = a^2 + \epsilon$ and $y = a^2 - \epsilon$. These lines intersect the graph of f at points with x -coordinates $\sqrt{a^2 - \epsilon}$ and $\sqrt{a^2 + \epsilon}$, as illustrated in the figure. If x is in the open interval $(\sqrt{a^2 - \epsilon}, \sqrt{a^2 + \epsilon})$, then

$$\sqrt{a^2 - \epsilon} < x < \sqrt{a^2 + \epsilon}.$$

Consequently,

$$a^2 - \epsilon < x^2 < a^2 + \epsilon,$$

that is, $f(x) = x^2$ is in the open interval $(a^2 - \epsilon, a^2 + \epsilon)$. Geometrically, this means that the point (x, x^2) on the graph of f lies between the horizontal lines. Let us choose a number δ smaller than both $\sqrt{a^2 + \epsilon} - a$ and $a - \sqrt{a^2 - \epsilon}$ as illustrated in Figure 2.8. It follows that if x is in the interval $(a - \delta, a + \delta)$, then x is also in $(\sqrt{a^2 - \epsilon}, \sqrt{a^2 + \epsilon})$, and, therefore, $f(x)$ is in $(a^2 - \epsilon, a^2 + \epsilon)$. Hence, by Definition (2.4), $\lim_{x \rightarrow a} x^2 = a^2$. Although we have considered only $a > 0$, a similar argument applies if $a \leq 0$. ■

To shorten explanations, whenever the notation $\lim_{x \rightarrow a} f(x) = L$ is used we shall often assume that all the conditions given in Definition (2.3) are satisfied. Thus, it may not always be pointed out that f is defined on an open interval containing a . Moreover, we shall not always specify L but merely write “ $\lim_{x \rightarrow a} f(x)$ exists,” or “ $f(x)$ has a limit as x approaches a .” The phrase “find $\lim_{x \rightarrow a} f(x)$ ” means “find a number L such that $\lim_{x \rightarrow a} f(x) = L$.” If no such L exists we write “ $\lim_{x \rightarrow a} f(x)$ does not exist.”

It can be proved that if $f(x)$ has a limit as x approaches a , then that limit is unique (see Appendix II).

In the following example we return to the function considered at the beginning of this section and *prove* that the limit exists by means of Definition (2.3).

Example 2 Prove that $\lim_{x \rightarrow 4} \frac{1}{2}(3x - 1) = \frac{11}{2}$.

Solution Let $f(x) = \frac{1}{2}(3x - 1)$, $a = 4$, and $L = \frac{11}{2}$. According to Definition (2.3) we must show that for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$\text{if } 0 < |x - 4| < \delta, \text{ then } \left| \frac{1}{2}(3x - 1) - \frac{11}{2} \right| < \varepsilon.$$

A clue to the choice of δ can be found by examining the last inequality involving ε . The following is a list of equivalent inequalities.

$$\begin{aligned} \left| \frac{1}{2}(3x - 1) - \frac{11}{2} \right| &< \varepsilon \\ \frac{1}{2}|(3x - 1) - 11| &< \varepsilon \\ |3x - 1 - 11| &< 2\varepsilon \\ |3x - 12| &< 2\varepsilon \\ 3|x - 4| &< 2\varepsilon \\ |x - 4| &< \frac{2}{3}\varepsilon \end{aligned}$$

The final inequality gives us the needed clue. If we let $\delta = \frac{2}{3}\varepsilon$, then if $0 < |x - 4| < \delta$, the last inequality in the list is true and consequently so is the first. Hence by Definition (2.3), $\lim_{x \rightarrow 4} \frac{1}{2}(3x - 1) = \frac{11}{2}$.

It is also possible to give a geometric proof similar to that used in Example 1. ■

It was relatively easy to use the definition of limit in the previous examples because $f(x)$ was a simple expression involving x . Limits of more complicated functions may also be verified by direct applications of the definition; however, the task of showing that for every $\varepsilon > 0$ there exists a suitable $\delta > 0$ often requires a great deal of ingenuity. In Section 2.3 we shall introduce theorems which can be used to find many limits without resorting to a search for the general number δ that appears in Definition (2.3).

The next two examples indicate how the geometric interpretation illustrated in Figure 2.7 may be used to show that certain limits do not exist.

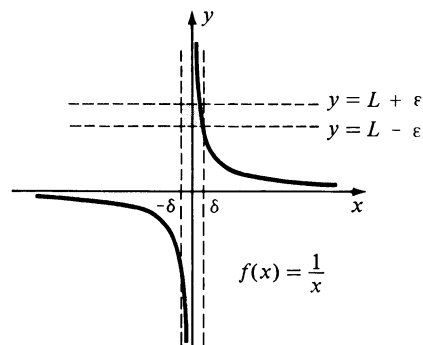


FIGURE 2.9

Example 3 Show that $\lim_{x \rightarrow 0} 1/x$ does not exist.

Solution Let us proceed in an indirect manner. Thus, suppose it were true that

$$\lim_{x \rightarrow 0} \frac{1}{x} = L$$

for some number L . Let us consider any pair of horizontal lines $y = L \pm \varepsilon$ as illustrated in Figure 2.9. Since we are assuming that the limit exists, it

should be possible to find an open interval $(0 - \delta, 0 + \delta)$ or equivalently, $(-\delta, \delta)$, containing 0, such that whenever $-\delta < x < \delta$ and $x \neq 0$, the point $(x, 1/x)$ on the graph lies between the horizontal lines. However, since $1/x$ can be made as large as desired by choosing x close to 0, not every point $(x, 1/x)$ with nonzero x -coordinate in $(-\delta, \delta)$ has this property. Consequently our supposition is false; that is, the limit does not exist. ■

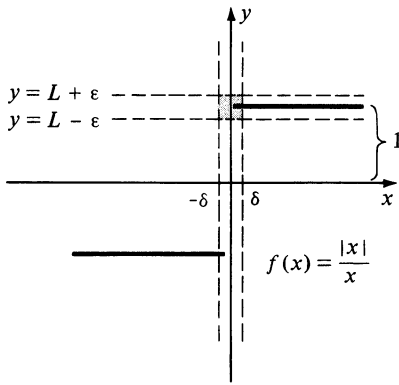


FIGURE 2.10

Example 4 If $f(x) = |x|/x$, show that $\lim_{x \rightarrow 0} f(x)$ does not exist.

Solution If $x > 0$, then $|x|/x = x/x = 1$ and hence, to the right of the y -axis, the graph of f coincides with the line $y = 1$. If $x < 0$, then $|x|/x = -x/x = -1$, which means that to the left of the y -axis the graph of f coincides with the line $y = -1$. If it were true that $\lim_{x \rightarrow 0} f(x) = L$ for some L , then the preceding remarks imply that $-1 \leq L \leq 1$. As shown in Figure 2.10, if we consider any pair of horizontal lines $y = L \pm \varepsilon$, where $0 < \varepsilon < 1$, then there exist points on the graph which are not between these lines for some nonzero x in every interval $(-\delta, \delta)$ containing 0. It follows that the limit does not exist. ■

2.2 Exercises

Establish the limits in Exercises 1–10 by means of Definition (2.3).

1 $\lim_{x \rightarrow 2} (5x - 3) = 7$

2 $\lim_{x \rightarrow -3} (2x + 1) = -5$

3 $\lim_{x \rightarrow -6} (10 - 9x) = 64$

4 $\lim_{x \rightarrow 4} (8x - 15) = 17$

5 $\lim_{x \rightarrow 5} \left(\frac{x}{4} + 2 \right) = \frac{13}{4}$

6 $\lim_{x \rightarrow 6} \left(9 - \frac{x}{6} \right) = 8$

7 $\lim_{x \rightarrow 3} 5 = 5$

8 $\lim_{x \rightarrow 5} 3 = 3$

9 $\lim_{x \rightarrow \pi} x = \pi$

10 $\lim_{x \rightarrow a} 2 = 2$

Use the method illustrated in Examples 3 and 4 to show that the limits in Exercises 11–14 do not exist.

11 $\lim_{x \rightarrow 3} \frac{|x - 3|}{x - 3}$

12 $\lim_{x \rightarrow -2} \frac{x + 2}{|x + 2|}$

13 $\lim_{x \rightarrow -5} \frac{1}{(x + 5)}$

14 $\lim_{x \rightarrow 1} \frac{1}{(x - 1)^2}$

15 Give an example of a function f which is defined at a and for which $\lim_{x \rightarrow a} f(x)$ exists, but $\lim_{x \rightarrow a} f(x) \neq f(a)$.

16 If f is the greatest integer function and a is any integer, show that $\lim_{x \rightarrow a} f(x)$ does not exist.

17 Let f be defined by the following conditions: $f(x) = 0$ if x is rational and $f(x) = 1$ if x is irrational. Prove that for every real number a , $\lim_{x \rightarrow a} f(x)$ does not exist.

18 Why is it impossible to investigate $\lim_{x \rightarrow 0} \sqrt{x}$ by means of Definition (2.3)?

Use the graphical technique illustrated in Example 1 to verify the limits in Exercises 19–22.

19 $\lim_{x \rightarrow a} x^3 = a^3$

20 $\lim_{x \rightarrow a} \sqrt{x} = \sqrt{a}, \quad a > 0$

21 $\lim_{x \rightarrow a} \sqrt[3]{x} = \sqrt[3]{a}$

22 $\lim_{x \rightarrow a} (x^2 + 1)^2 = (a^2 + 1)^2$

2.3 Theorems on Limits

It would be an excruciating task to solve each problem on limits by means of Definition (2.3). The purpose of this section is to introduce theorems that may be used to simplify the process. To prove the theorems it is necessary to employ the definition of limit; however, once they are established it will be possible to determine many limits without referring to an ε or a δ . Several theorems are proved in this section; the remaining proofs may be found in Appendix II.

The simplest limit to consider involves the constant function defined by $f(x) = c$, where c is a real number. If f is represented geometrically by means of coordinate lines l and l' , then *every* arrow from l terminates at the same point on l' , namely, the point with coordinate c , as indicated in Figure 2.11.

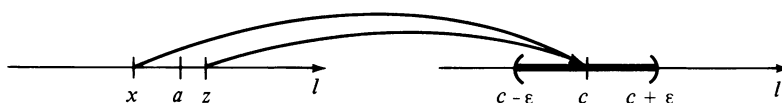


FIGURE 2.11

It is easy to prove that for every real number a , $\lim_{x \rightarrow a} f(x) = c$. Thus, if $\varepsilon > 0$, consider the open interval $(c - \varepsilon, c + \varepsilon)$ on l' as illustrated in the figure. Since $f(x) = c$ is always in this interval, *any* number δ will satisfy the conditions of Definition (2.3); that is, for *every* $\delta > 0$,

$$\text{if } 0 < |x - a| < \delta, \text{ then } |f(x) - c| < \varepsilon.$$

We have proved the following theorem.

Theorem (2.5)

If a and c are any real numbers, then $\lim_{x \rightarrow a} c = c$.

Sometimes Theorem (2.5) is phrased “the limit of a constant is the constant.” To illustrate,

$$\lim_{x \rightarrow 3} 8 = 8, \quad \lim_{x \rightarrow 8} 3 = 3, \quad \lim_{x \rightarrow \pi} \sqrt{2} = \sqrt{2}, \quad \text{and} \quad \lim_{x \rightarrow a} 0 = 0.$$

The next result tells us that in order to find the limit of a linear function as x approaches a , we merely substitute a for x .

Theorem (2.6)

If a , b , and m are real numbers, then $\lim_{x \rightarrow a} (mx + b) = ma + b$.

Proof If $m = 0$, then $mx + b = b$ and the statement of the theorem reduces to $\lim_{x \rightarrow a} b = b$, which was proved in Theorem (2.5).

Next suppose $m \neq 0$. If we let $f(x) = mx + b$ and $L = ma + b$, then according to Definition (2.3) we must show that for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$\text{if } 0 < |x - a| < \delta, \text{ then } |(mx + b) - (ma + b)| < \varepsilon.$$

As in the solution of Example 2 in the previous section, a clue to the choice of δ can be found by examining the inequality involving ε . All inequalities in the following list are equivalent.

$$\begin{aligned} |(mx + b) - (ma + b)| &< \varepsilon \\ |mx - ma| &< \varepsilon \\ |m||x - a| &< \varepsilon \\ |x - a| &< \frac{\varepsilon}{|m|} \end{aligned}$$

The last inequality suggests that we choose $\delta = \varepsilon/|m|$. Thus, given any $\varepsilon > 0$,

$$\text{if } 0 < |x - a| < \delta, \text{ where } \delta = \varepsilon/|m|$$

then the last inequality in the list is true, and hence, so is the first inequality, which is what we wished to prove. $\square$

As special cases of Theorem (2.6), we have

$$\begin{aligned} \lim_{x \rightarrow a} x &= a \\ \lim_{x \rightarrow 4} (3x - 5) &= 3 \cdot 4 - 5 = 7 \\ \lim_{x \rightarrow \sqrt{2}} (13x + \sqrt{2}) &= 13\sqrt{2} + \sqrt{2} = 14\sqrt{2}. \end{aligned}$$

The next theorem states that if a function f has a positive limit as x approaches a , then $f(x)$ is positive throughout some open interval containing a , with the possible exception of a .

Theorem (2.7)

If $\lim_{x \rightarrow a} f(x) = L$ and $L > 0$, then there exists an open interval $(a - \delta, a + \delta)$ containing a such that $f(x) > 0$ for all x in $(a - \delta, a + \delta)$, except possibly $x = a$.

Proof If $\varepsilon = L/2$, then the interval $(L - \varepsilon, L + \varepsilon)$ contains only positive numbers. By Definition (2.4) there exists a $\delta > 0$ such that whenever x is in the open interval $(a - \delta, a + \delta)$ and $x \neq a$, then $f(x)$ is in $(L - \varepsilon, L + \varepsilon)$, and hence $f(x) > 0$. $\square$

In like manner, it can be shown that if f has a *negative* limit as x approaches a , then there is an open interval I containing a such that $f(x) < 0$ for every x in I , with the possible exception of $x = a$.

Many functions may be expressed as sums, differences, products, and quotients of other functions. In particular, suppose a function s is a sum of two functions f and g , so that $s(x) = f(x) + g(x)$ for every x in the domain of s . If $f(x)$ and $g(x)$ have limits L and M , respectively, as x approaches a , it is natural to conclude that $s(x)$ has the limit $L + M$ as x approaches a . The fact that this and analogous statements hold for products and quotients are consequences of the next theorem.

Theorem (2.8)

If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then

- (i) $\lim_{x \rightarrow a} [f(x) + g(x)] = L + M.$
- (ii) $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = L \cdot M.$
- (iii) $\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{L}{M},$ provided $M \neq 0.$
- (iv) $\lim_{x \rightarrow a} [cf(x)] = cL.$
- (v) $\lim_{x \rightarrow a} [f(x) - g(x)] = L - M.$

Although the conclusions of Theorem (2.8) appear to be intuitively evident, the proofs are rather technical and require some ingenuity. Proofs for (i)–(iii) may be found in Appendix II. Part (iv) of the theorem follows readily from part (ii) and Theorem (2.5) as follows:

$$\lim_{x \rightarrow a} [cf(x)] = \left[\lim_{x \rightarrow a} c \right] \left[\lim_{x \rightarrow a} f(x) \right] = cL.$$

Finally, to prove (v) we may write

$$f(x) - g(x) = f(x) + (-1)g(x)$$

and then use (i) and (iv) (with $c = -1$).

The conclusions of Theorem (2.8) are often written as follows:

- (i) $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$
- (ii) $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$
- (iii) $\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)},$ if $\lim_{x \rightarrow a} g(x) \neq 0$
- (iv) $\lim_{x \rightarrow a} [cf(x)] = c \left[\lim_{x \rightarrow a} f(x) \right]$
- (v) $\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x).$

Example 1 Find $\lim_{x \rightarrow 2} \frac{3x + 4}{5x + 7}$.

Solution The numerator and denominator of the indicated quotient define linear functions whose limits exist by Theorem (2.6). Moreover, the limit of the denominator is not 0. Hence by (iii) of Theorem (2.8) and (2.6),

$$\lim_{x \rightarrow 2} \frac{3x + 4}{5x + 7} = \frac{\lim_{x \rightarrow 2} (3x + 4)}{\lim_{x \rightarrow 2} (5x + 7)} = \frac{3(2) + 4}{5(2) + 7} = \frac{10}{17}. \quad \blacksquare$$

Notice the simple manner in which the limit in Example 1 was found. It would be a lengthy task to verify the limit by means of Definition (2.3).

Theorem (2.8) may be extended to limits of sums, differences, products, and quotients that involve any number of functions. An application of (ii) to three (equal) functions is given in the next example.

Example 2 Prove that for every real number a , $\lim_{x \rightarrow a} x^3 = a^3$.

Solution Since $\lim_{x \rightarrow a} x = a$ we may write

$$\begin{aligned} \lim_{x \rightarrow a} x^3 &= \lim_{x \rightarrow a} (x \cdot x \cdot x) \\ &= \left(\lim_{x \rightarrow a} x \right) \cdot \left(\lim_{x \rightarrow a} x \right) \cdot \left(\lim_{x \rightarrow a} x \right) \\ &= a \cdot a \cdot a = a^3. \quad \blacksquare \end{aligned}$$

The technique used in Example 2 can be extended to x^n , where n is any positive integer. We merely write x^n as a product $x \cdot x \cdot \dots \cdot x$ of n factors and then take the limit of each factor. This gives us (i) of the next theorem. Part (ii) may be proved in similar fashion by using (ii) of Theorem (2.8).

Theorem (2.9)

If n is a positive integer, then

- (i) $\lim_{x \rightarrow a} x^n = a^n$.
- (ii) $\lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n$, provided $\lim_{x \rightarrow a} f(x)$ exists.

Example 3 Find $\lim_{x \rightarrow 2} (3x + 4)^5$.

Solution Applying (ii) of (2.9) and Theorem (2.6),

$$\begin{aligned} \lim_{x \rightarrow 2} (3x + 4)^5 &= \left[\lim_{x \rightarrow 2} (3x + 4) \right]^5 \\ &= [3(2) + 4]^5 \\ &= 10^5 = 100,000. \quad \blacksquare \end{aligned}$$

Example 4 Find $\lim_{x \rightarrow -2} (5x^3 + 3x^2 - 6)$.

Solution We may proceed as follows (supply reasons):

$$\begin{aligned}
 \lim_{x \rightarrow -2} (5x^3 + 3x^2 - 6) &= \lim_{x \rightarrow -2} (5x^3) + \lim_{x \rightarrow -2} (3x^2) - \lim_{x \rightarrow -2} (6) \\
 &= 5 \lim_{x \rightarrow -2} (x^3) + 3 \lim_{x \rightarrow -2} (x^2) - 6 \\
 &= 5(-2)^3 + 3(-2)^2 - 6 \\
 &= 5(-8) + 3(4) - 6 = -34.
 \end{aligned}$$

■

Note that the limit in Example 4 is the number obtained by substituting -2 for x in $5x^3 + 3x^2 - 6$. The next theorem states that the same is true for the limit of *every* polynomial.

Theorem (2.10)

If f is a polynomial function and a is a real number, then

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Proof We may write $f(x)$ in the form

$$f(x) = b_n x^n + b_{n-1} x^{n-1} + \cdots + b_0$$

where the b_i are real numbers. As in Example 4,

$$\begin{aligned}
 \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} (b_n x^n) + \lim_{x \rightarrow a} (b_{n-1} x^{n-1}) + \cdots + \lim_{x \rightarrow a} b_0 \\
 &= b_n \lim_{x \rightarrow a} (x^n) + b_{n-1} \lim_{x \rightarrow a} (x^{n-1}) + \cdots + \lim_{x \rightarrow a} b_0 \\
 &= b_n a^n + b_{n-1} a^{n-1} + \cdots + b_0 = f(a).
 \end{aligned}$$

□

Corollary (2.11)

If q is a rational function and a is in the domain of q , then

$$\lim_{x \rightarrow a} q(x) = q(a).$$

Proof We may write $q(x) = f(x)/h(x)$ where f and h are polynomial functions. If a is in the domain of q , then $h(a) \neq 0$. Using (iii) of Theorem (2.8) and (2.10),

$$\lim_{x \rightarrow a} q(x) = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} h(x)} = \frac{f(a)}{h(a)} = q(a).$$

□

Example 5 Find $\lim_{x \rightarrow 3} \frac{5x^2 - 2x + 1}{6x - 7}$.

Solution Applying Corollary (2.11),

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{5x^2 - 2x + 1}{6x - 7} &= \frac{5(3)^2 - 2(3) + 1}{6(3) - 7} \\ &= \frac{45 - 6 + 1}{18 - 7} = \frac{40}{11}. \end{aligned}$$

The following theorem states that for positive integral roots of x , we may determine a limit by substitution. The proof makes use of the definition of limit and may be found in Appendix II.

Theorem (2.12)

If $a > 0$ and n is a positive integer, or if $a \leq 0$ and n is an odd positive integer, then

$$\lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a}.$$

If m and n are positive integers and $a > 0$, then using (ii) of (2.9) and Theorem (2.12),

$$\lim_{x \rightarrow a} (\sqrt[n]{x})^m = \left(\lim_{x \rightarrow a} \sqrt[n]{x} \right)^m = (\sqrt[n]{a})^m.$$

In terms of rational exponents,

$$\lim_{x \rightarrow a} x^{m/n} = a^{m/n}.$$

This limit formula may be extended to negative exponents by writing $x^{-r} = 1/x^r$ and then using (iii) of (2.8).

Example 6 Find $\lim_{x \rightarrow 8} \frac{x^{2/3} + 3\sqrt{x}}{4 - (16/x)}$.

Solution The reader should supply reasons for each of the following steps.

$$\begin{aligned} \lim_{x \rightarrow 8} \frac{x^{2/3} + 3\sqrt{x}}{4 - (16/x)} &= \frac{\lim_{x \rightarrow 8} (x^{2/3} + 3\sqrt{x})}{\lim_{x \rightarrow 8} (4 - (16/x))} \\ &= \frac{\lim_{x \rightarrow 8} x^{2/3} + \lim_{x \rightarrow 8} 3\sqrt{x}}{\lim_{x \rightarrow 8} 4 - \lim_{x \rightarrow 8} (16/x)} \\ &= \frac{8^{2/3} + 3\sqrt{8}}{4 - (16/8)} \\ &= \frac{4 + 6\sqrt{2}}{4 - 2} = 2 + 3\sqrt{2} \end{aligned}$$

Theorem (2.13)

If a function f has a limit as x approaches a , then

$$\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}$$

provided either n is an odd positive integer or n is an even positive integer and $\lim_{x \rightarrow a} f(x) > 0$.

The preceding theorem will be proved in Section 2.6. In the meantime we shall use it whenever applicable to gain experience in finding limits that involve roots of algebraic expressions.

Example 7 Find $\lim_{x \rightarrow 5} \sqrt[3]{3x^2 - 4x + 9}$.

Solution Using Theorems (2.13) and (2.10),

$$\begin{aligned} \lim_{x \rightarrow 5} \sqrt[3]{3x^2 - 4x + 9} &= \sqrt[3]{\lim_{x \rightarrow 5} (3x^2 - 4x + 9)} \\ &= \sqrt[3]{75 - 20 + 9} = \sqrt[3]{64} = 4. \end{aligned}$$

The beginning student should not be misled by the preceding examples. It is not always possible to find limits merely by substitution. Sometimes other devices must be employed. The next theorem concerns three functions f , h , and g , where $h(x)$ is always “sandwiched” between $f(x)$ and $g(x)$. If f and g have a common limit L as x approaches a , then as stated below, h must have the same limit.

The Sandwich Theorem (2.14)

If $f(x) \leq h(x) \leq g(x)$ for all x in an open interval containing a , except possibly at a , and

$$\text{if} \quad \lim_{x \rightarrow a} f(x) = L = \lim_{x \rightarrow a} g(x),$$

$$\text{then} \quad \lim_{x \rightarrow a} h(x) = L.$$

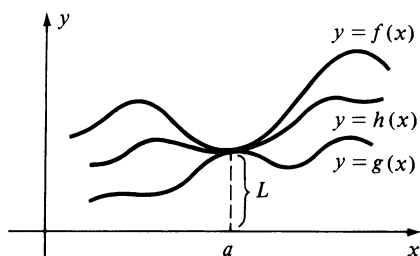


FIGURE 2.12

A proof of the Sandwich Theorem based on the definition of limit may be found in Appendix II. The result is also clear from geometric considerations. Specifically, if $f(x) \leq h(x) \leq g(x)$ for all x in an open interval containing x , then the graph of h lies “between” the graphs of f and g in that interval, as illustrated in Figure 2.12. If f and g have the same limit L as x approaches a , then evidently, h also has the limit L .

Example 8 Prove that $\lim_{x \rightarrow 0} x \sin(1/x) = 0$.

Solution The limit cannot be found by substituting 0 for x , or by using an algebraic manipulation. However, since all values of the sine function are between -1 and 1 it follows that if $x \neq 0$, $|\sin(1/x)| \leq 1$ and, therefore,

$$\left| x \sin \frac{1}{x} \right| = |x| \left| \sin \frac{1}{x} \right| \leq |x|.$$

Consequently,
$$0 \leq \left| x \sin \frac{1}{x} \right| \leq |x|.$$

It is not difficult to show that $\lim_{x \rightarrow 0} |x| = 0$. Hence, by the Sandwich Theorem (2.14), with $f(x) = 0$ and $g(x) = |x|$, we see that

$$\lim_{x \rightarrow 0} \left| x \sin \frac{1}{x} \right| = 0.$$

It now follows from the Definition of Limit (2.3) that

$$\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0. \quad \blacksquare$$

2.3 Exercises

In Exercises 1–36 find the limits, if they exist.

1 $\lim_{x \rightarrow -2} (3x^3 - 2x + 7)$

2 $\lim_{x \rightarrow 4} (5x^2 - 9x - 8)$

3 $\lim_{x \rightarrow \sqrt{2}} (x^2 + 3)(x - 4)$

4 $\lim_{t \rightarrow -3} (3t + 4)(7t - 9)$

5 $\lim_{x \rightarrow 4} \sqrt[3]{x^2 - 5x - 4}$

6 $\lim_{x \rightarrow -2} \sqrt{x^4 - 4x + 1}$

7 $\lim_{x \rightarrow 7} 0$

8 $\lim_{x \rightarrow 1/2} \frac{4x^2 - 6x + 3}{16x^3 + 8x - 7}$

9 $\lim_{x \rightarrow \sqrt{2}} 15$

10 $\lim_{x \rightarrow 15} \sqrt{2}$

11 $\lim_{x \rightarrow 1/2} \frac{2x^2 + 5x - 3}{6x^2 - 7x + 2}$

12 $\lim_{x \rightarrow -3} \frac{x + 3}{(1/x) + (1/3)}$

13 $\lim_{x \rightarrow 2} \frac{x - 2}{x^3 - 8}$

14 $\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{(x - 2)^2}$

15 $\lim_{x \rightarrow 16} \frac{x - 16}{\sqrt{x} - 4}$

16 $\lim_{x \rightarrow -2} \frac{x^3 + 8}{x^4 - 16}$

17 $\lim_{s \rightarrow 4} \frac{6s - 1}{2s - 9}$

18 $\lim_{x \rightarrow \pi} (x - 3.1416)$

19 $\lim_{x \rightarrow 1} \left(\frac{x^2}{x - 1} - \frac{1}{x - 1} \right)$

20 $\lim_{x \rightarrow 1} \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^6$

21 $\lim_{x \rightarrow 16} \frac{2\sqrt{x} + x^{3/2}}{\sqrt[4]{x} + 5}$

22 $\lim_{x \rightarrow -8} \frac{16x^{2/3}}{4 - x^{4/3}}$

23 $\lim_{x \rightarrow 3} \sqrt[3]{\frac{2 + 5x - 3x^3}{x^2 - 1}}$

24 $\lim_{x \rightarrow \pi} \sqrt[5]{\frac{x - \pi}{x + \pi}}$

25 $\lim_{h \rightarrow 0} \frac{4 - \sqrt{16 + h}}{h}$

26 $\lim_{h \rightarrow 0} \left(\frac{1}{h} \right) \left(\frac{1}{\sqrt{1 + h}} - 1 \right)$

27 $\lim_{x \rightarrow 1} \frac{(x - 1)^5}{x^5 - 1}$

28 $\lim_{x \rightarrow 6} (x + 4)^3(x - 6)^2$

29 $\lim_{v \rightarrow 3} v^2(3v - 4)(9 - v^3)$

30 $\lim_{k \rightarrow 2} \sqrt{3k^2 + 4} \sqrt[3]{3k + 2}$

31 $\lim_{t \rightarrow -1} \frac{(4t^2 + 5t - 3)^3}{(6t + 5)^4}$

32 $\lim_{t \rightarrow 7} \frac{\sqrt[5]{3 - 5t}}{(t - 5)^3}$

33 $\lim_{x \rightarrow 9} \frac{x^2 - 81}{3 - \sqrt{x}}$

34 $\lim_{x \rightarrow 8} \frac{x - 8}{\sqrt[3]{x} - 2}$

35 $\lim_{h \rightarrow 0} \frac{(2 + h)^{-2} - 2^{-2}}{h}$

36 $\lim_{h \rightarrow 0} \frac{(9 + h)^{-1} - 9^{-1}}{h}$

37 If r is any rational number and $a > 0$, prove that $\lim_{x \rightarrow a} x^r = a^r$. Under what conditions will this be true if $a < 0$?

- 38 If $\lim_{x \rightarrow a} f(x) = L \neq 0$ and $\lim_{x \rightarrow a} g(x) = 0$, prove that $\lim_{x \rightarrow a} [f(x)/g(x)]$ does not exist. (Hint: Assume there is a number M such that $\lim_{x \rightarrow a} [f(x)/g(x)] = M$ and consider

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} \left[g(x) \cdot \frac{f(x)}{g(x)} \right].$$

- 39 Use the Sandwich Theorem and the fact that $\lim_{x \rightarrow 0} (|x| + 1) = 1$ to prove that $\lim_{x \rightarrow 0} (x^2 + 1) = 1$.
- 40 Use the Sandwich Theorem with $f(x) = 0$ and $g(x) = |x|$ to prove that

$$\lim_{x \rightarrow 0} \frac{|x|}{\sqrt{x^4 + 4x^2 + 7}} = 0.$$

- 41 If c is a nonnegative real number and $0 \leq f(x) \leq c$ for every x , prove that $\lim_{x \rightarrow 0} x^2 f(x) = 0$.
- 42 Prove that $\lim_{x \rightarrow 0} x^4 \sin(1/\sqrt[3]{x}) = 0$. (Hint: See Example 8.)
- 43 Prove that if f has a negative limit as x approaches a , then there is some open interval I containing a such that $f(x)$ is negative for every x in I except possibly $x = a$.

2.4 One-Sided Limits

If $f(x) = \sqrt{x - 2}$ and $a > 2$, then f is defined throughout an open interval containing a and, by Theorem (2.13),

$$\lim_{x \rightarrow a} \sqrt{x - 2} = \sqrt{\lim_{x \rightarrow a} (x - 2)} = \sqrt{a - 2}.$$

The case $a = 2$ is not covered by Definition (2.3) since there is no open interval containing 2 throughout which f is defined (note that $\sqrt{x - 2}$ is not real if $x < 2$). A natural way to extend the definition of limit to include this exceptional case is to restrict x to values *greater* than 2. Thus, we replace the condition $2 - \delta < x < 2 + \delta$, which arises from Definition (2.3), by the condition $2 < x < 2 + \delta$. The corresponding limit is called *the limit of $f(x)$ as x approaches 2 from the right*, or the *right-hand limit* of $\sqrt{x - 2}$ as x approaches 2.

Definition (2.15)

Let f be a function that is defined on an open interval (a, c) , and let L be a real number. The statement

$$\lim_{x \rightarrow a^+} f(x) = L$$

means that for every $\varepsilon > 0$, there exists $\delta > 0$, such that

$$\text{if } a < x < a + \delta, \text{ then } |f(x) - L| < \varepsilon.$$

If $\lim_{x \rightarrow a^+} f(x) = L$, we say that **the limit of $f(x)$, as x approaches a from the right, is L** . We also refer to L as the **right-hand limit** of $f(x)$ as x approaches a . The symbol $x \rightarrow a^+$ is used to indicate that values of x are always larger than

a. Note that the only difference between Definitions (2.15) and (2.3) is that for *right-hand limits* we restrict x to the *right half* $(a, a + \delta)$ of the interval $(a - \delta, a + \delta)$. Definition (2.15) is illustrated geometrically in (i) of Figure 2.13. Intuitively, we think of $f(x)$ getting close to L as x gets close to a , through values *larger* than a .

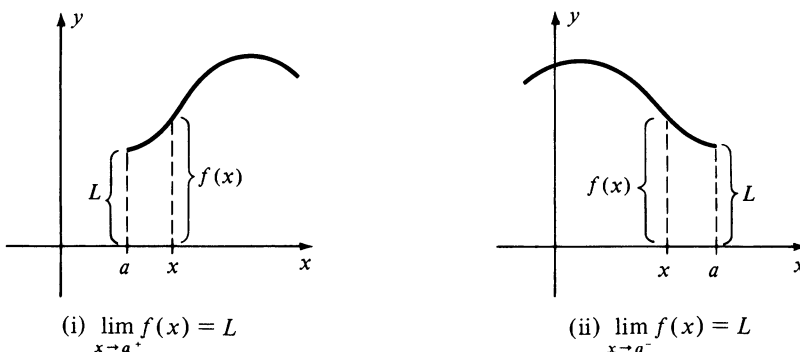


FIGURE 2.13

The notion of *left-hand limit* is defined in similar fashion. For example, if $f(x) = \sqrt{2 - x}$, then we restrict x to values *less* than 2. The general definition follows.

Definition (2.16)

Let f be a function that is defined on an open interval (c, a) , and let L be a real number. The statement

$$\lim_{x \rightarrow a^-} f(x) = L$$

means that for every $\varepsilon > 0$, there exists a $\delta > 0$, such that

$$\text{if } a - \delta < x < a, \text{ then } |f(x) - L| < \varepsilon.$$

If $\lim_{x \rightarrow a^-} f(x) = L$, we say that **the limit of $f(x)$ as x approaches a from the left, is L** ; or that L is the **left-hand limit** of $f(x)$ as x approaches a . The symbol $x \rightarrow a^-$ is used to indicate that x is restricted to values *less* than a . A geometric illustration of Definition (2.16) is given in (ii) of Figure 2.13. Note that for the *left-hand limit*, x is in the *left half* $(a - \delta, a)$ of the interval $(a - \delta, a + \delta)$.

Sometimes Definitions (2.15) and (2.16) are referred to as **one-sided limits** of $f(x)$ as x approaches a . The relation between one-sided limits and limits is stated in the next theorem. The proof is left as an exercise.

Theorem (2.17)

If f is defined throughout an open interval containing a , except possibly at a itself, then $\lim_{x \rightarrow a} f(x) = L$ if and only if both $\lim_{x \rightarrow a^-} f(x) = L$ and $\lim_{x \rightarrow a^+} f(x) = L$.

The preceding theorem tells us that the limit of $f(x)$ as x approaches a exists if and only if both the right- and left-hand limits exist and are equal.

Theorems similar to the limit theorems of the previous section can be proved for one-sided limits. For example,

$$\lim_{x \rightarrow a^+} [f(x) + g(x)] = \lim_{x \rightarrow a^+} f(x) + \lim_{x \rightarrow a^+} g(x)$$

and

$$\lim_{x \rightarrow a^+} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a^+} f(x)}$$

with the usual restrictions on the existence of limits and n th roots. Analogous results are true for left-hand limits.

Example 1 Find $\lim_{x \rightarrow 2^+} (1 + \sqrt{x - 2})$.

Solution Using (one-sided) limit theorems,

$$\begin{aligned} \lim_{x \rightarrow 2^+} (1 + \sqrt{x - 2}) &= \lim_{x \rightarrow 2^+} 1 + \lim_{x \rightarrow 2^+} \sqrt{x - 2} \\ &= 1 + \sqrt{\lim_{x \rightarrow 2^+} (x - 2)} \\ &= 1 + 0 = 1. \end{aligned}$$

The graph of $f(x) = 1 + \sqrt{x - 2}$ is sketched in Figure 2.14. Note that there is no left-hand limit, since $\sqrt{x - 2}$ is not a real number if $x < 2$. ■

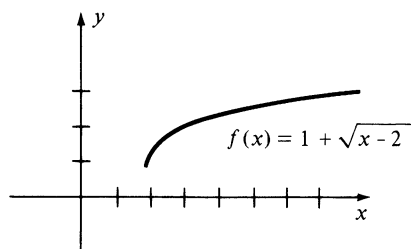


FIGURE 2.14

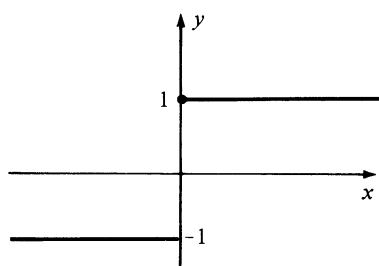


FIGURE 2.15

Example 2 Suppose $f(x) = |x|/x$ if $x \neq 0$ and $f(0) = 1$. Find $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow 0^-} f(x)$. What is $\lim_{x \rightarrow 0} f(x)$?

Solution If $x > 0$, then $|x| = x$ and $f(x) = x/x = 1$. Consequently,

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} 1 = 1.$$

If $x < 0$, then $|x| = -x$ and $f(x) = -x/x = -1$. Therefore,

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (-1) = -1.$$

Since these right- and left-hand limits are unequal, it follows from Theorem (2.17) that $\lim_{x \rightarrow 0} f(x)$ does not exist. The graph of f is sketched in Figure 2.15. ■

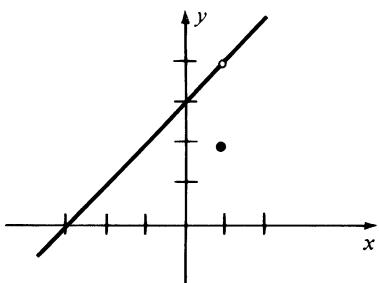


FIGURE 2.16

Example 3 Suppose $f(x) = x + 3$ if $x \neq 1$ and $f(1) = 2$. Find $\lim_{x \rightarrow 1^-} f(x)$, $\lim_{x \rightarrow 1^+} f(x)$, and $\lim_{x \rightarrow 1} f(x)$.

Solution The graph of f consists of the point $P(1, 2)$ and all points on the line $y = x + 3$ except the point with coordinates $(1, 4)$ as shown in Figure 2.16. Evidently, $\lim_{x \rightarrow 1^+} f(x) = 4 = \lim_{x \rightarrow 1^-} f(x)$. Hence by Theorem (2.17), $\lim_{x \rightarrow 1} f(x) = 4$. Note that $\lim_{x \rightarrow 1} f(x) \neq f(1)$. ■

2.4 Exercises

In Exercises 1–22, find the limits, if they exist.

- 1 (a) $\lim_{x \rightarrow 5^-} \sqrt{5-x}$
- (b) $\lim_{x \rightarrow 5^+} \sqrt{5-x}$
- (c) $\lim_{x \rightarrow 5} \sqrt{5-x}$
- 3 (a) $\lim_{x \rightarrow 1^-} \sqrt[3]{x^3-1}$
- (b) $\lim_{x \rightarrow 1^+} \sqrt[3]{x^3-1}$
- (c) $\lim_{x \rightarrow 1} \sqrt[3]{x^3-1}$
- 5 (a) $\lim_{x \rightarrow 4^-} \frac{|x-4|}{x-4}$
- (b) $\lim_{x \rightarrow 4^+} \frac{|x-4|}{x-4}$
- (c) $\lim_{x \rightarrow 4} \frac{|x-4|}{x-4}$
- 7 $\lim_{x \rightarrow 0^+} (4 + \sqrt{x})$
- 9 $\lim_{x \rightarrow -6^+} (\sqrt{x+6} + x)$
- 11 $\lim_{x \rightarrow 5^+} (\sqrt{x^2-25} + 3)$
- 13 $\lim_{x \rightarrow 2^+} \frac{4-x^2}{2-x}$
- 15 $\lim_{x \rightarrow 3^+} \frac{\sqrt{(x-3)^2}}{x-3}$
- 17 $\lim_{x \rightarrow 5^+} \frac{1 + \sqrt{2x-10}}{x+3}$
- 19 $\lim_{x \rightarrow -7^+} \frac{x+7}{|x+7|}$
- 21 $\lim_{x \rightarrow 0^+} \frac{1}{x}$
- 2 (a) $\lim_{x \rightarrow 2^-} \sqrt{8-x^3}$
- (b) $\lim_{x \rightarrow 2^+} \sqrt{8-x^3}$
- (c) $\lim_{x \rightarrow 2} \sqrt{8-x^3}$
- 4 (a) $\lim_{x \rightarrow -8^+} x^{2/3}$
- (b) $\lim_{x \rightarrow -8^-} x^{2/3}$
- (c) $\lim_{x \rightarrow -8} x^{2/3}$
- 6 (a) $\lim_{x \rightarrow -5^+} \frac{x+5}{|x+5|}$
- (b) $\lim_{x \rightarrow -5^-} \frac{x+5}{|x+5|}$
- (c) $\lim_{x \rightarrow -5} \frac{x+5}{|x+5|}$
- 8 $\lim_{x \rightarrow 0^+} (4x^{3/2} - \sqrt{x} + 3)$
- 10 $\lim_{x \rightarrow 5/2^-} (\sqrt{5-2x} - x^2)$
- 12 $\lim_{x \rightarrow 3^-} x\sqrt{9-x^2}$
- 14 $\lim_{x \rightarrow -4^+} \frac{2x^2+5x-12}{x^2+3x-4}$
- 16 $\lim_{x \rightarrow -10^-} \frac{x+10}{\sqrt{(x+10)^2}}$
- 18 $\lim_{x \rightarrow 4^+} \frac{\sqrt[4]{x^2-16}}{x+4}$
- 20 $\lim_{x \rightarrow \pi^-} \frac{|\pi-x|}{x-\pi}$
- 22 $\lim_{x \rightarrow 8^-} \frac{1}{x-8}$

For each f defined in Exercises 23 and 24, find $\lim_{x \rightarrow 2^-} f(x)$, $\lim_{x \rightarrow 2^+} f(x)$, and sketch the graph of f .

- 23 $f(x) = \begin{cases} 3x & \text{if } x \leq 2 \\ x^2 & \text{if } x > 2 \end{cases}$
- 24 $f(x) = \begin{cases} x^3 & \text{if } x \leq 2 \\ 4-2x & \text{if } x > 2 \end{cases}$

In Exercises 25 and 26 find $\lim_{x \rightarrow -3^+} f(x)$, $\lim_{x \rightarrow -3^-} f(x)$, and $\lim_{x \rightarrow -3} f(x)$, if they exist.

- 25 $f(x) = \begin{cases} 1/(2-3x) & \text{if } x < -3 \\ \sqrt[3]{x+2} & \text{if } x \geq -3 \end{cases}$
- 26 $f(x) = \begin{cases} 9/x^2 & \text{if } x \leq -3 \\ 4+x & \text{if } x > -3 \end{cases}$

In Exercises 27–29, n denotes an arbitrary integer. For each function f , sketch the graph of f and find $\lim_{x \rightarrow n^-} f(x)$ and $\lim_{x \rightarrow n^+} f(x)$.

- 27 $f(x) = (-1)^n$ if $n \leq x < n+1$
- 28 $f(x) = \begin{cases} 0 & \text{if } x = n \\ 1 & \text{if } x \neq n \end{cases}$
- 29 $f(x) = \begin{cases} x & \text{if } x = n \\ 0 & \text{if } x \neq n \end{cases}$

In Exercises 30 and 31, $[]$ denotes the greatest integer function and n is an arbitrary integer.

- 30 Find $\lim_{x \rightarrow n^-} [x]$ and $\lim_{x \rightarrow n^+} [x]$.
- 31 If $f(x) = x - [x]$, find $\lim_{x \rightarrow n^-} f(x)$ and $\lim_{x \rightarrow n^+} f(x)$.
- 32 If $f(x) = (x^2-9)/(x-3)$ for $x \neq 3$ and $f(3) = 5$, find $\lim_{x \rightarrow 3^+} f(x)$, $\lim_{x \rightarrow 3^-} f(x)$, and $\lim_{x \rightarrow 3} f(x)$.
- 33 If f is a polynomial function, prove that $\lim_{x \rightarrow a^+} f(x) = f(a)$ and $\lim_{x \rightarrow a^-} f(x) = f(a)$ for every real number a .
- 34 Prove Theorem (2.17).
- 35 Sketch geometric interpretations for right-hand limits which are analogous to those in Figures 2.5 and 2.6. Do the same for left-hand limits.

2.5 Limits of Trigonometric Functions

Whenever we discuss limits of trigonometric expressions involving $\sin t$, $\cos x$, $\tan \theta$, etc., we shall assume that each variable represents a real number or the radian measure of an angle. The following result is important for future developments.

Theorem (2.18)

$$\lim_{t \rightarrow 0} \sin t = 0$$

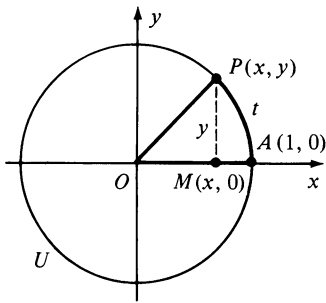


FIGURE 2.17

Proof Let us first prove that $\lim_{t \rightarrow 0^+} \sin t = 0$. Since we are only interested in positive values of t near zero, there is no loss of generality in assuming that $0 < t < \pi/2$. Let U be the circle of radius 1 with center at the origin of a rectangular coordinate system, and let A be the point $(1, 0)$. If, as illustrated in Figure 2.17, $P(x, y)$ is the point on U that corresponds to t , then by (1.26) the radian measure of angle AOP is t .

Referring to the figure we see that

$$0 < y < t$$

or, since $y = \sin t$,

$$0 < \sin t < t.$$

Since $\lim_{t \rightarrow 0^+} t = 0$, it follows from the Sandwich Theorem (2.14) that $\lim_{t \rightarrow 0^+} \sin t = 0$.

To complete the proof it is sufficient to show that $\lim_{t \rightarrow 0^-} \sin t = 0$. If $-\pi/2 < t < 0$, then $0 < -t < \pi/2$ and hence, from the first part of the proof,

$$0 < \sin(-t) < -t.$$

Multiplying the last inequality by -1 and using the fact that $\sin(-t) = -\sin t$ gives us

$$t < \sin t < 0.$$

Since $\lim_{t \rightarrow 0^-} t = 0$, it follows from the Sandwich Theorem that $\lim_{t \rightarrow 0^-} \sin t = 0$. $\square$

Corollary (2.19)

$$\lim_{t \rightarrow 0} \cos t = 1$$

Proof Since $\sin^2 t + \cos^2 t = 1$, it follows that $\cos t = \pm \sqrt{1 - \sin^2 t}$. If $-\pi/2 < t < \pi/2$, then $\cos t$ is positive, and hence $\cos t = \sqrt{1 - \sin^2 t}$. Consequently,

$$\begin{aligned} \lim_{t \rightarrow 0} \cos t &= \lim_{t \rightarrow 0} \sqrt{1 - \sin^2 t} = \sqrt{\lim_{t \rightarrow 0} (1 - \sin^2 t)} \\ &= \sqrt{1 - 0} = 1. \end{aligned}$$

$\square$

For our work in Section 3.4 it will be essential to know the limits of $(\sin t)/t$ and $(1 - \cos t)/t$ as t approaches 0. These are established in Theorems (2.21) and (2.22). In the proof of (2.21) we shall make use of the following result.

Theorem (2.20)

If θ is the radian measure of a central angle of a circle of radius r , then the area A of the sector determined by θ is

$$A = \frac{1}{2}r^2\theta.$$

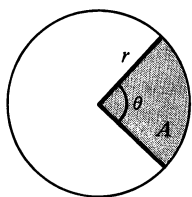


FIGURE 2.18

Proof A typical central angle θ and the sector it determines are shown in Figure 2.18. The area of the sector is directly proportional to θ , that is,

$$A = k\theta$$

for some real number k . For example, the area determined by an angle of 2 radians is twice the area determined by an angle of 1 radian. In particular, if $\theta = 2\pi$, then the sector is the entire circle, and $A = \pi r^2$. Thus

$$\pi r^2 = k(2\pi), \quad \text{or} \quad k = \frac{1}{2}r^2$$

and therefore

$$A = \frac{1}{2}r^2\theta. \quad \square$$

Theorem (2.21)

$$\lim_{t \rightarrow 0} \frac{\sin t}{t} = 1$$

Proof If $0 < t < \pi/2$ we have the situation illustrated in Figure 2.19 where U is a unit circle.

If A_1 is the area of triangle AOP , A_2 the area of circular sector AOP , and A_3 the area of triangle AOQ , then

$$A_1 < A_2 < A_3.$$

Using the formula for the area of a triangle and Theorem (2.20) we obtain

$$A_1 = \frac{1}{2}(1)d(M, P) = \frac{1}{2}y = \frac{1}{2}\sin t$$

$$A_2 = \frac{1}{2}(1)^2t = \frac{1}{2}t$$

$$A_3 = \frac{1}{2}(1)d(A, Q) = \frac{1}{2}\tan t.$$

Thus

$$\frac{1}{2}\sin t < \frac{1}{2}t < \frac{1}{2}\tan t.$$

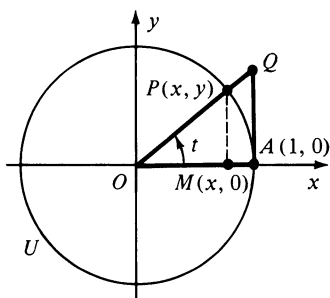


FIGURE 2.19

Dividing by $\frac{1}{2} \sin t$ and using the fact that $\tan t = \sin t / \cos t$ gives us

$$1 < \frac{t}{\sin t} < \frac{1}{\cos t}$$

or, equivalently,

$$(*) \quad 1 > \frac{\sin t}{t} > \cos t.$$

If $-\pi/2 < t < 0$, then $0 < -t < \pi/2$, and from the result just established,

$$1 > \frac{\sin(-t)}{-t} > \cos(-t).$$

Since $\sin(-t) = -\sin t$ and $\cos(-t) = \cos t$, this inequality reduces to (*). This shows that (*) is also true if $-\pi/2 < t < 0$, and hence is true for every t in the open interval $(-\pi/2, \pi/2)$ except $t = 0$. Since $\lim_{t \rightarrow 0} \cos t = 1$, and $(\sin t)/t$ is always between $\cos t$ and 1, it follows from the Sandwich Theorem that

$$\lim_{t \rightarrow 0} \frac{\sin t}{t} = 1. \quad \square$$

Roughly speaking, Theorem (2.21) implies that if t is close to 0, then $(\sin t)/t$ is close to 1. Another way of stating this is to write $\sin t \approx t$ for small values of t . It is important to remember that if t denotes an angle, then *radian measure must be used in Theorem (2.21)* and in the approximation formula $\sin t \approx t$. To illustrate, trigonometric tables or a calculator show that to five decimal places,

$$\sin(0.06) \approx 0.05996$$

$$\sin(0.05) \approx 0.04998$$

$$\sin(0.04) \approx 0.03999$$

$$\sin(0.03) \approx 0.03000.$$

Theorem (2.22)

$$\lim_{t \rightarrow 0} \frac{1 - \cos t}{t} = 0$$

Proof

We may change the form of $(1 - \cos t)/t$ as follows:

$$\begin{aligned} \frac{1 - \cos t}{t} &= \frac{1 - \cos t}{t} \cdot \frac{1 + \cos t}{1 + \cos t} \\ &= \frac{1 - \cos^2 t}{t(1 + \cos t)} \\ &= \frac{\sin^2 t}{t(1 + \cos t)} \\ &= \frac{\sin t}{t} \cdot \frac{\sin t}{1 + \cos t}. \end{aligned}$$

Consequently,

$$\begin{aligned}\lim_{t \rightarrow 0} \frac{1 - \cos t}{t} &= \lim_{t \rightarrow 0} \left(\frac{\sin t}{t} \cdot \frac{\sin t}{1 + \cos t} \right) \\ &= \left(\lim_{t \rightarrow 0} \frac{\sin t}{t} \right) \left(\lim_{t \rightarrow 0} \frac{\sin t}{1 + \cos t} \right) \\ &= 1 \cdot \left(\frac{0}{1 + 1} \right) = 1 \cdot 0 = 0.\end{aligned}$$

□

Example 1 Find $\lim_{x \rightarrow 0} (\sin 5x)/2x$.

Solution We cannot apply Theorem (2.21) directly, since the given expression is not in the form $(\sin t)/t$. However, we may introduce this form (with $t = 5x$) by using the following algebraic manipulation:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin 5x}{2x} &= \lim_{x \rightarrow 0} \frac{1}{2} \frac{\sin 5x}{x} \\ &= \lim_{x \rightarrow 0} \frac{5}{2} \frac{\sin 5x}{5x} \\ &= \frac{5}{2} \lim_{x \rightarrow 0} \frac{\sin 5x}{5x}.\end{aligned}$$

It follows from the definition of limit that $x \rightarrow 0$ may be replaced by $5x \rightarrow 0$. Hence, by Theorem (2.21), with $t = 5x$, we see that

$$\lim_{x \rightarrow 0} \frac{\sin 5x}{2x} = \frac{5}{2} (1) = \frac{5}{2}.$$

Warning: When working problems of this type, remember that $\sin 5x \neq 5 \sin x$. ■

Example 2 Find $\lim_{t \rightarrow 0} (\tan t)/2t$.

Solution Using the fact that $\tan t = \sin t/\cos t$,

$$\begin{aligned}\lim_{t \rightarrow 0} \frac{\tan t}{2t} &= \lim_{t \rightarrow 0} \left(\frac{1}{2} \cdot \frac{\sin t}{t} \cdot \frac{1}{\cos t} \right) \\ &= \frac{1}{2} \cdot 1 \cdot 1 = \frac{1}{2}.\end{aligned}$$

■

Example 3 Find $\lim_{x \rightarrow 0} (2x + 1 - \cos x)/3x$.

Solution We plan to use Theorem (2.22). With this in mind we begin by isolating the part of the quotient that involves $(1 - \cos x)/x$ and then proceed as follows.

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{2x + 1 - \cos x}{3x} &= \lim_{x \rightarrow 0} \left(\frac{2x}{3x} + \frac{1 - \cos x}{3x} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{2x}{3x} \right) + \lim_{x \rightarrow 0} \frac{1}{3} \left(\frac{1 - \cos x}{x} \right) \\ &= \lim_{x \rightarrow 0} \frac{2}{3} + \frac{1}{3} \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} \\ &= \frac{2}{3} + \frac{1}{3} \cdot 0 = \frac{2}{3}\end{aligned}$$

■

2.5 Exercises

Find the limits in Exercises 1–26.

$$1 \quad \lim_{x \rightarrow 0} \frac{x}{\sin x}$$

$$2 \quad \lim_{x \rightarrow 0} \frac{\sin x}{\sqrt[3]{x}}$$

$$3 \quad \lim_{t \rightarrow 0} \frac{\sin^3 t}{(2t)^3}$$

$$4 \quad \lim_{\theta \rightarrow 0} \frac{3\theta + \sin \theta}{\theta}$$

$$5 \quad \lim_{x \rightarrow 0} \frac{2 + \sin x}{3 + x}$$

$$6 \quad \lim_{t \rightarrow 0} \frac{1 - \cos 3t}{t}$$

$$7 \quad \lim_{\theta \rightarrow 0} \frac{2 \cos \theta - 2}{3\theta}$$

$$8 \quad \lim_{x \rightarrow 0} \frac{x^2 + 1}{x + \cos x}$$

$$9 \quad \lim_{x \rightarrow 0} \frac{\sin(-3x)}{4x}$$

$$10 \quad \lim_{x \rightarrow 0} \frac{x \sin x}{x^2 + 1}$$

$$11 \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^{2/3}}$$

$$12 \quad \lim_{x \rightarrow 0} \frac{1 - 2x^2 - 2 \cos x + \cos^2 x}{x^2}$$

$$13 \quad \lim_{t \rightarrow 0} \frac{4t^2 + 3t \sin t}{t^2}$$

$$14 \quad \lim_{x \rightarrow 0} \frac{x \cos x - x^2}{2x}$$

$$15 \quad \lim_{t \rightarrow 0} \frac{\cos t}{1 - \sin t}$$

$$16 \quad \lim_{t \rightarrow 0} \frac{\sin t}{1 + \cos t}$$

$$17 \quad \lim_{t \rightarrow 0} \frac{1 - \cos t}{\sin t}$$

$$18 \quad \lim_{x \rightarrow 0} \frac{\sin(x/2)}{x}$$

$$19 \quad \lim_{x \rightarrow 0} \frac{x + \tan x}{\sin x}$$

$$20 \quad \lim_{t \rightarrow 0} \frac{\sin^2 2t}{t^2}$$

$$21 \quad \lim_{x \rightarrow 0} x \cot x$$

$$22 \quad \lim_{x \rightarrow 0} \frac{\csc 2x}{\cot x}$$

$$23 \quad \lim_{\alpha \rightarrow 0} \alpha^2 \csc^2 \alpha$$

$$24 \quad \lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 5x}$$

$$25 \quad \lim_{v \rightarrow 0} \frac{\cos(v + \pi/2)}{v}$$

$$26 \quad \lim_{x \rightarrow 0} \frac{\sin^2(x/2)}{\sin x}$$

Establish the limits in Exercises 27–30, where a and b are any nonzero real numbers.

$$27 \quad \lim_{x \rightarrow 0} \frac{\sin ax}{bx} = \frac{a}{b}$$

$$28 \quad \lim_{x \rightarrow 0} \frac{1 - \cos ax}{bx} = 0$$

$$29 \quad \lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx} = \frac{a}{b}$$

$$30 \quad \lim_{x \rightarrow 0} \frac{\cos ax}{\cos bx} = 1$$

2.6 Continuous Functions

In arriving at the definition of $\lim_{x \rightarrow a} f(x)$ we emphasized the restriction $x \neq a$. Several examples in preceding sections have brought out the fact that $\lim_{x \rightarrow a} f(x)$ may exist even though f is undefined at a . Let us now turn our attention to the case in which a is in the domain of f . If f is defined at a and $\lim_{x \rightarrow a} f(x)$ exists, then this limit may, or may not, equal $f(a)$. If $\lim_{x \rightarrow a} f(x) = f(a)$ then f is said to be *continuous* at a according to the next definition.

Definition (2.23)

A function f is **continuous** at a number a if the following three conditions are satisfied.

- (i) f is defined on an open interval containing a .
- (ii) $\lim_{x \rightarrow a} f(x)$ exists.
- (iii) $\lim_{x \rightarrow a} f(x) = f(a)$.

If f is not continuous at a , then we say it is **discontinuous** at a , or has a **discontinuity** at a .

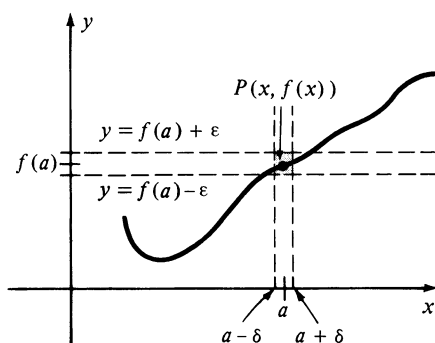


FIGURE 2.20

If f is continuous at a , then by (i) of Definition (2.23) there is a point $(a, f(a))$ on the graph of f . Moreover, since $\lim_{x \rightarrow a} f(x) = f(a)$, the closer x is to a , the closer $f(x)$ is to $f(a)$ or, in geometric terms, the closer the point $(x, f(x))$ on the graph of f is to the point $(a, f(a))$. More precisely, as illustrated in Figure 2.20, for every pair of horizontal lines $y = f(a) \pm \varepsilon$, there exist vertical lines $x = a \pm \delta$, such that if $a - \delta < x < a + \delta$, then the point $(x, f(x))$ on the graph of f lies within the shaded rectangular region.

Functions that are continuous at every number in a given interval are sometimes thought of as functions whose graphs can be sketched without lifting the pencil from the paper; that is, there are no breaks in the graph. Another interpretation of a continuous function f is that a small change in x produces only a small change in the functional value $f(x)$. These are not accurate descriptions, but rather devices to help develop an intuitive feeling for continuous functions.

Example 1

- (a) Prove that the sine function is continuous at 0.
- (b) Prove that a polynomial function is continuous at every real number a .
- (c) Prove that a rational function is continuous at every real number in its domain.

Solution

(a) We must verify the three conditions of Definition (2.23). Since $\sin x$ is defined for every x , it is defined in an open interval containing 0, and consequently (i) of (2.23) is fulfilled. Moreover, by Theorem (2.18),

$$\lim_{x \rightarrow 0} \sin x = 0$$

and hence condition (ii) of (2.23) holds. Finally, since $\sin 0 = 0$ we have

$$\lim_{x \rightarrow 0} \sin x = \sin 0$$

and therefore condition (iii) is true. This completes the proof.

(b) A polynomial function f is defined throughout $\mathbb{R}$. Moreover, by Theorem (2.10), $\lim_{x \rightarrow a} f(x) = f(a)$ for every real number a . Thus f satisfies conditions (i)–(iii) of Definition (2.23) and hence is continuous at a .

(c) If q is a rational function, then $q = f/h$, where f and h are polynomial functions. Consequently q is defined for all real numbers *except* the zeros of h . It follows that if $h(a) \neq 0$, then q is defined throughout an open interval containing a . Moreover, by (2.11), $\lim_{x \rightarrow a} q(x) = q(a)$. Applying Definition (2.23), q is continuous at a . ■

Since the notion of continuity involves the fact that $\lim_{x \rightarrow a} f(x) = f(a)$, the following result may be obtained by replacing L in Definition (2.3) by $f(a)$.

Theorem (2.24)

If a function f is defined on an open interval containing a , then f is continuous at a if for every $\varepsilon > 0$, there exists a $\delta > 0$, such that

$$\text{if } |x - a| < \delta, \text{ then } |f(x) - f(a)| < \varepsilon.$$

Note that we do not require $0 < |x - a|$ in Theorem (2.24), since f is defined at a and, moreover, if $x = a$, then

$$|f(x) - f(a)| = |f(a) - f(a)| = 0 < \varepsilon.$$

Graphs of several functions that are *not* continuous at a are sketched in Figure 2.21. The function having the graph illustrated in (i) fails to be continuous since f is undefined at a . The functions whose graphs are sketched in (ii) and (iii) are discontinuous at a since $\lim_{x \rightarrow a} f(x)$ does not exist. For the function with a graph as in (iv), both $f(a)$ and $\lim_{x \rightarrow a} f(x)$ exist but they are unequal and hence f is discontinuous at a by (iii) of Definition (2.23). The last illustration shows the necessity for checking all three conditions of the definition.

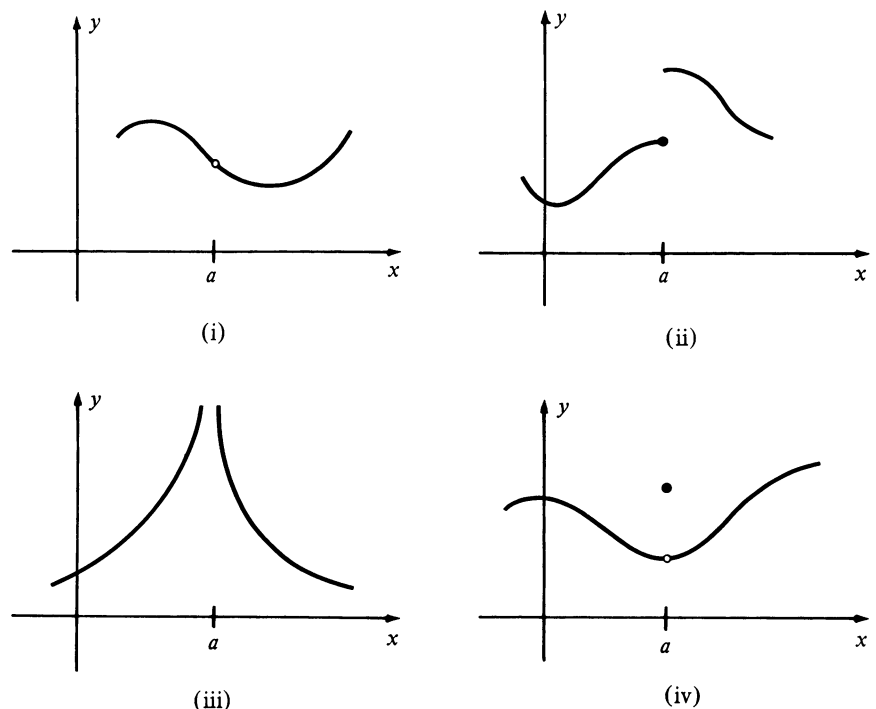


FIGURE 2.21

As a specific illustration of (iv) in Figure 2.21, consider the function f of Example 3 in Section 2.4, where $f(x) = x + 3$ if $x \neq 1$ and $f(1) = 2$ (see Figure 2.16). Since

$$\lim_{x \rightarrow 1} f(x) = 4 \neq f(1),$$

condition (iii) of Definition (2.23) is not satisfied, and hence f has a discontinuity at $x = 1$.

If $f(x) = 1/x$, then f has a discontinuity at $x = 0$. In this case *none* of the conditions of Definition (2.23) is satisfied (see Figure 2.9).

The functions whose graphs are sketched in Figure 2.21 appear to be continuous at numbers other than a . Most functions considered in calculus are of this type; that is, they may be discontinuous at certain numbers of their domains and continuous elsewhere.

If a function f is continuous at every number in an open interval (a, b) , we say that f is **continuous on the interval (a, b)** . Similarly, a function is said to be continuous on an infinite interval of the form (a, ∞) or $(-\infty, b)$ if it is continuous at every number in the interval. The next definition covers the case of a closed interval.

Definition (2.25)

Let a function f be defined on a closed interval $[a, b]$. The function f is **continuous on $[a, b]$** if it is continuous on (a, b) and if, in addition,

$$\lim_{x \rightarrow a^+} f(x) = f(a) \quad \text{and} \quad \lim_{x \rightarrow b^-} f(x) = f(b).$$

If a function f has either a right-hand or left-hand limit of the type indicated in Definition (2.25) we say that f is **continuous from the right at a** or that f is **continuous from the left at b** , respectively.

Example 2 If $f(x) = \sqrt{9 - x^2}$, sketch the graph of f and prove that f is continuous on the closed interval $[-3, 3]$.

Solution By (1.13), the graph of $x^2 + y^2 = 9$ or equivalently $y^2 = 9 - x^2$ is a circle with center at the origin and radius 3. It follows that the graph of $y = \sqrt{9 - x^2}$ and, therefore, the graph of f , is the upper half of that circle (see Figure 2.22).

If $-3 < c < 3$ then, using Theorem (2.13),

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} \sqrt{9 - x^2} = \sqrt{9 - c^2} = f(c).$$

Hence, by Definition (2.23), f is continuous at c .

According to Definition (2.25), all that remains is to check the endpoints of the interval using one-sided limits. Since

$$\lim_{x \rightarrow -3^+} f(x) = \lim_{x \rightarrow -3^+} \sqrt{9 - x^2} = \sqrt{9 - 9} = 0 = f(-3),$$

f is continuous from the right at -3 . We also have

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} \sqrt{9 - x^2} = \sqrt{9 - 9} = 0 = f(3)$$

and hence f is continuous from the left at 3. This completes the proof that f is continuous on $[-3, 3]$. ■

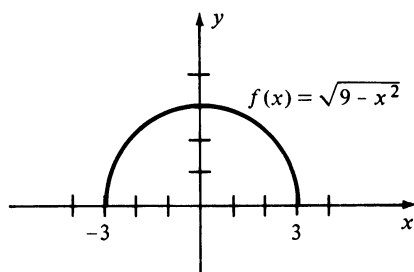


FIGURE 2.22

It should be evident how to define continuity on other types of intervals. For example, a function f is said to be continuous on $[a, b)$ or $[a, \infty)$ if it is continuous at every number greater than a in the interval and if, in addition, f is continuous from the right at a . For intervals of the form $(a, b]$ or $(-\infty, b]$ we require continuity at every number less than b in the interval and also continuity from the left at b .

As illustrated in the next example, when asked to discuss the continuity of a function f we shall list the largest intervals on which f is continuous. Of course f will also be continuous on any subinterval of those intervals.

Example 3 Discuss the continuity of f if $f(x) = \sqrt{x^2 - 9}/(x - 4)$.

Solution The function is undefined if the denominator $x - 4$ is zero (that is, if $x = 4$) or if the radicand $x^2 - 9$ is negative (that is, if $-3 < x < 3$). Any other real number is in one of the intervals $(-\infty, -3]$, $[3, 4)$, or $(4, \infty)$. The proof that f is continuous on each of these intervals is similar to that given in the solution of Example 2. Thus to prove continuity on $[3, 4)$ it is necessary to show that

$$\lim_{x \rightarrow c} f(x) = f(c) \quad \text{if } 3 < c < 4$$

and also that

$$\lim_{x \rightarrow 3^+} f(x) = f(3).$$

We shall leave the details of the proof for this, and the other intervals, to the reader. ■

Limit theorems discussed in Section 2.3 may be used to establish the following important theorem.

Theorem (2.26)

If the functions f and g are continuous at a , then so are the sum $f + g$, the difference $f - g$, the product fg , and, if $g(a) \neq 0$, the quotient f/g .

Proof If f and g are continuous at a , then $\lim_{x \rightarrow a} f(x) = f(a)$ and $\lim_{x \rightarrow a} g(x) = g(a)$. By the definition of sum $(f + g)(x) = f(x) + g(x)$. Consequently,

$$\begin{aligned} \lim_{x \rightarrow a} (f + g)(x) &= \lim_{x \rightarrow a} [f(x) + g(x)] \\ &= \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) \\ &= f(a) + g(a) \\ &= (f + g)(a). \end{aligned}$$

This proves that $f + g$ is continuous at a . The remainder of the theorem is proved in similar fashion. □

If f and g are continuous on an interval I it follows that $f + g$, $f - g$, and fg are continuous on I . If, in addition, $g(a) \neq 0$ throughout I , then f/g is continuous on I . These results may be extended to more than two functions; that is, sums, differences, products, or quotients involving any number of continuous functions are continuous (provided zero denominators do not occur).

The next result on limits of composite functions has many applications.

Theorem (2.27)

If f and g are functions such that $\lim_{x \rightarrow a} g(x) = b$, and if f is continuous at b , then

$$\lim_{x \rightarrow a} f(g(x)) = f(b) = f\left(\lim_{x \rightarrow a} g(x)\right).$$

Proof As was pointed out in Chapter 1 (see Figure 1.34), the composite function $f(g(x))$ may be represented geometrically by means of three real lines l , l' , and l'' as shown in Figure 2.23, where to each coordinate x on l there corresponds the coordinate $g(x)$ on l' and then, in turn, $f(g(x))$ on l'' . We wish to prove that $f(g(x))$ has the limit $f(b)$ as x approaches a . In terms of Definition (2.3) we must show that for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$(i) \quad \text{if } 0 < |x - a| < \delta, \text{ then } |f(g(x)) - f(b)| < \varepsilon.$$

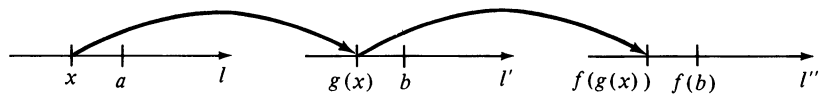


FIGURE 2.23

Let us begin by considering the interval $(f(b) - \varepsilon, f(b) + \varepsilon)$ on l'' shown in color in Figure 2.24. Since f is continuous at b , $\lim_{z \rightarrow b} f(z) = f(b)$ and hence, as illustrated in the figure, there exists a number $\delta_1 > 0$ such that

$$(ii) \quad \text{if } |z - b| < \delta_1, \text{ then } |f(z) - f(b)| < \varepsilon.$$

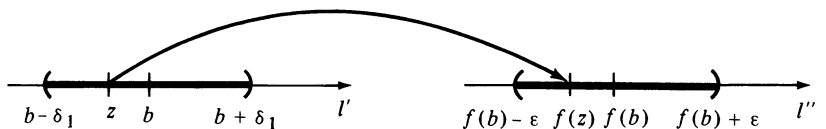


FIGURE 2.24

In particular, if we let $z = g(x)$ in (ii) it follows that

$$(iii) \quad \text{if } |g(x) - b| < \delta_1, \text{ then } |f(g(x)) - f(b)| < \varepsilon.$$

Next, turning our attention to the interval $(b - \delta_1, b + \delta_1)$ on l' and using the definition of $\lim_{x \rightarrow a} g(x) = b$, we obtain the fact illustrated in Figure 2.25, that there exists a $\delta > 0$ such that

$$(iv) \quad \text{if } 0 < |x - a| < \delta, \text{ then } |g(x) - b| < \delta_1.$$

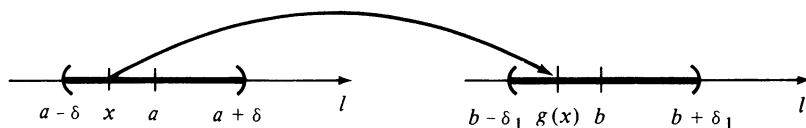


FIGURE 2.25

Finally, combining (iv) and (iii) we see that

$$\text{if } 0 < |x - a| < \delta \text{ then } |f(g(x)) - f(b)| < \varepsilon$$

which is the desired conclusion (i). $\square$

The principal use of Theorem (2.27) is to establish other theorems. To illustrate, if n is a positive integer and $f(x) = \sqrt[n]{x}$, then

$$f(g(x)) = \sqrt[n]{g(x)}$$

and

$$f\left(\lim_{x \rightarrow a} g(x)\right) = \sqrt[n]{\lim_{x \rightarrow a} g(x)}.$$

If we now use the fact that

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right)$$

the result stated in Theorem (2.13) is obtained, that is,

$$\lim_{x \rightarrow a} \sqrt[n]{g(x)} = \sqrt[n]{\lim_{x \rightarrow a} g(x)},$$

where it is assumed that the indicated n th roots exist.

The next result follows directly from Theorem 2.27.

Theorem (2.28)

If g is continuous at a and f is continuous at $b = g(a)$, then

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right) = f(g(a)).$$

The preceding theorem states that the composite function of f by g is continuous at a . This result may be extended to functions that are continuous on intervals. Sometimes this is expressed by the statement “the composite function of a continuous function by a continuous function is continuous.”

Example 4 If $f(x) = |x|$, prove that f is continuous at every real number a .

Solution Since $|x| = \sqrt{x^2}$ we have, by (2.13) and (2.9),

$$\begin{aligned}\lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} |x| = \lim_{x \rightarrow a} \sqrt{x^2} \\ &= \sqrt{\lim_{x \rightarrow a} x^2} = \sqrt{a^2} = |a| = f(a).\end{aligned}$$

Hence, from Definition (2.23), f is continuous at a . ■

A proof of the following important property of continuous functions may be found in more advanced texts on calculus.

The Intermediate Value Theorem
(2.29)

If a function f is continuous on a closed interval $[a, b]$ and if $f(a) \neq f(b)$, then f takes on every value between $f(a)$ and $f(b)$ in the interval $[a, b]$.

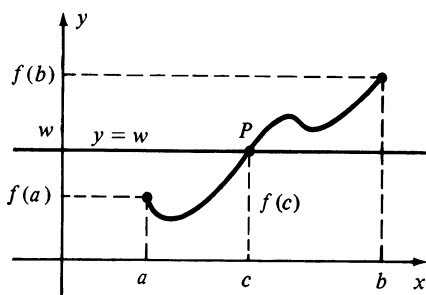


FIGURE 2.26

Theorem (2.29) states that if w is any number between $f(a)$ and $f(b)$, then there is a number c between a and b such that $f(c) = w$. If the graph of the continuous function f is regarded as extending in an unbroken manner from the point $(a, f(a))$ to the point $(b, f(b))$, as illustrated in Figure 2.26, then for any number w between $f(a)$ and $f(b)$ it appears that a horizontal line with y -intercept w should intersect the graph in at least one point P . The x -coordinate c of P is a number such that $f(c) = w$.

Example 5 Verify the Intermediate Value Theorem (2.29) if $f(x) = \sqrt{x+1}$ and the interval is $[3, 24]$.

Solution The function f is continuous on $[3, 24]$ and $f(3) = 2$, $f(24) = 5$. If w is any real number between 2 and 5 we must find a number c in the interval $[3, 24]$ such that $f(c) = w$, that is, $\sqrt{c+1} = w$. Squaring both sides of the last equation and solving for c we obtain $c = w^2 - 1$. This number c is in the interval $[3, 24]$, since if $2 < w < 5$, then

$$4 < w^2 < 25, \quad \text{or} \quad 3 < w^2 - 1 < 24.$$

To check our work we see that

$$f(c) = f(w^2 - 1) = \sqrt{(w^2 - 1) + 1} = w. \quad \blacksquare$$

A corollary of Theorem (2.29) is that if $f(a)$ and $f(b)$ have opposite signs, then there is a number c between a and b such that $f(c) = 0$; that is, f has a zero at c . Geometrically, this implies that if the point $(a, f(a))$ on the graph of a continuous function lies below the x -axis, and the point $(b, f(b))$ lies above the x -axis, or vice versa, then the graph crosses the x -axis at some point $(c, 0)$, where $a < c < b$.

A useful consequence of the Intermediate Value Theorem is the following, where the interval referred to may be either closed, open, half-open, or infinite.

Theorem (2.30)

If a function f is continuous and has no zeros on an interval, then either $f(x) > 0$ or $f(x) < 0$ for all x in the interval.

Proof The conclusion of the theorem states that under the given hypothesis, $f(x)$ has the same sign throughout the interval. If this conclusion were *false*, then there would exist numbers x_1 and x_2 in the interval such that $f(x_1) > 0$ and $f(x_2) < 0$. By our preceding remarks this, in turn, would imply that $f(c) = 0$ for some number c between x_1 and x_2 , contrary to hypothesis. Thus, the conclusion must be true. $\square$

2.6 Exercises

In Exercises 1–12 show that the function f is continuous at the given number a .

- 1 $f(x) = \sqrt{2x - 5} + 3x, a = 4$
- 2 $f(x) = 3x^2 + 7 - 1/\sqrt{-x}, a = -2$
- 3 $f(x) = x/(x^2 - 4), a = 3$
- 4 $f(x) = 1/x, a = 10^{-6}$
- 5 $f(x) = \sqrt[3]{x^2} + 2, a = -5$
- 6 $f(x) = \sqrt[3]{x}/(2x + 1), a = 8$
- 7 $f(x) = \cos x, a = 0$ (Hint: See Corollary (2.19).)
- 8 $f(x) = \frac{\sin x + \cos x}{x^2 + 3x + 1}, a = 0$ (Hint: See Example 1 and Exercise 7.)
- 9 $f(x) = \sec x, a = 0$
- 10 $f(x) = \tan x, a = 0$
- 11 $f(x) = (\cos x)/(1 + \sin x), a = 0$
- 12 $f(x) = \sqrt{\frac{1 - \sin x}{1 + \cos x}}, a = 0$

In Exercises 13–16, show that f is continuous on the indicated interval.

- 13 $f(x) = \sqrt{x - 4}; [4, 8]$
- 14 $f(x) = \sqrt{16 - x}; (-\infty, 16]$

$$15 \quad f(x) = \frac{1}{x^2}; (0, \infty)$$

$$16 \quad f(x) = \frac{1}{x - 1}; (1, 3)$$

In Exercises 17–28 find all numbers for which the function f is continuous.

$$17 \quad f(x) = \frac{3x - 5}{2x^2 - x - 3}$$

$$18 \quad f(x) = \frac{x^2 - 9}{x - 3}$$

$$19 \quad f(x) = \sqrt{2x - 3} + x^2$$

$$20 \quad f(x) = \frac{x}{\sqrt[3]{x - 4}}$$

$$21 \quad f(x) = \frac{x - 1}{\sqrt{x^2 - 1}}$$

$$22 \quad f(x) = \frac{x}{\sqrt{1 - x^2}}$$

$$23 \quad f(x) = \frac{|x + 9|}{x + 9}$$

$$24 \quad f(x) = \frac{x}{x^2 + 1}$$

$$25 \quad f(x) = \frac{5}{x^3 - x^2}$$

$$26 \quad f(x) = \frac{4x - 7}{(x + 3)(x^2 + 2x - 8)}$$

$$27 \quad f(x) = \frac{\sqrt{x^2 - 9}\sqrt{25 - x^2}}{x - 4}$$

$$28 \quad f(x) = \frac{\sqrt{9 - x}}{\sqrt{x - 6}}$$

29–34 Discuss the discontinuities of the functions defined in Exercises 27–32 of Section 2.4.

$$35 \quad \text{Suppose } f(x) = \begin{cases} cx^2 - 3 & \text{if } x \leq 2 \\ cx + 2 & \text{if } x > 2 \end{cases}$$

Find a value of c such that f is continuous on $\mathbb{R}$.

$$36 \quad \text{Suppose } f(x) = \begin{cases} c^2x & \text{if } x < 1 \\ 3cx - 2 & \text{if } x \geq 1 \end{cases}$$

Determine all values of c such that f is continuous on $\mathbb{R}$.

$$37 \quad \text{Suppose } f(x) = \begin{cases} c & \text{if } x = -3 \\ \frac{9 - x^2}{4 - \sqrt{x^2 + 7}} & \text{if } |x| < 3 \\ d & \text{if } x = 3 \end{cases}$$

Find values of c and d such that f is continuous on $[-3, 3]$.

$$38 \quad \text{Suppose } f(x) = \begin{cases} 4x & \text{if } x \leq -1 \\ cx + d & \text{if } -1 < x \leq 2 \\ -5x & \text{if } x \geq 2 \end{cases}$$

Find values of c and d such that $f(x)$ is continuous on $\mathbb{R}$.

39 Suppose

$$f(x) = x^2 \quad \text{and} \quad g(x) = \begin{cases} -4 & \text{if } x \leq 0 \\ |x - 4| & \text{if } x > 0 \end{cases}$$

Determine whether the composite functions $f \circ g$ and $g \circ f$ are continuous at 0.

40 Let $f(x) = (x - [x])^2$ where $[\]$ denotes the greatest integer function. If n is any integer, show that (a) f is continuous on the interval $[n, n + 1)$; and (b) f is not continuous on $[n, n + 1]$. Sketch the graph of f .

41 Prove that if $f(x) = 1/x$, then f is continuous on every open interval that does not contain the origin. What is true for open intervals containing the origin?

In Exercises 42 and 43, is f continuous at 3? Explain.

$$42 \quad f(x) = \begin{cases} 1 & \text{if } x \neq 3 \\ 0 & \text{if } x = 3 \end{cases}$$

$$43 \quad f(x) = \begin{cases} \frac{|x - 3|}{x - 3} & \text{if } x \neq 3 \\ 1 & \text{if } x = 3 \end{cases}$$

In Exercises 44 and 45, is f continuous at 0? Explain.

$$44 \quad f(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

$$45 \quad f(x) = \begin{cases} \frac{1 - \cos x}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

46 Suppose $f(x) = 0$ if x is rational and $f(x) = 1$ if x is irrational. Prove that f is discontinuous at every real number a .

47 Prove that a function f is continuous at a if and only if $\lim_{h \rightarrow 0} f(a + h) = f(a)$.

48 If f is continuous on an interval containing c , and if $f(c) > 0$, prove that $f(x)$ is positive throughout an interval containing c . (Hint: See Theorem (2.17).)

In Exercises 49–52, verify the Intermediate Value Theorem (2.29) for f on the stated interval $[a, b]$ by showing that if w is any number between $f(a)$ and $f(b)$, then there is a number c in $[a, b]$ such that $f(c) = w$.

$$49 \quad f(x) = x^3 + 1; [-1, 2]$$

$$50 \quad f(x) = -x^3; [0, 2]$$

$$51 \quad f(x) = x^2 + 4x + 4; [0, 1]$$

$$52 \quad f(x) = x^2 - x; [-1, 3]$$

53 If $f(x) = x^3 - 5x^2 + 7x - 9$, use the Intermediate Value Theorem (2.29) to prove that there is a real number a such that $f(a) = 100$.

54 Prove that the equation $x^5 - 3x^4 - 2x^3 - x + 1 = 0$ has a solution between 0 and 1.

2.7 Review: Concepts

Define or discuss each of the following.

- 1 The limit of a function as x approaches a
- 2 The geometric interpretations of $\lim_{x \rightarrow a} f(x) = L$
- 3 Theorems on limits
- 4 Limits of polynomial and rational functions
- 5 Right- and left-hand limits of functions
- 6 Limits of trigonometric functions
- 7 Continuous function
- 8 Discontinuities of a function
- 9 Continuity on an interval
- 10 The Intermediate Value Theorem

Review: Exercises

In Exercises 1–26 find the limit, if it exists.

- 1 $\lim_{x \rightarrow 3} \frac{5x + 11}{\sqrt{x} + 1}$
- 2 $\lim_{x \rightarrow -2} \frac{6 - 7x}{(3 + 2x)^4}$
- 3 $\lim_{x \rightarrow -2} (2x - \sqrt{4x^2 + x})$
- 4 $\lim_{x \rightarrow 4^-} (x - \sqrt{16 - x^2})$
- 5 $\lim_{x \rightarrow 3/2} \frac{2x^2 + x - 6}{4x^2 - 4x - 3}$
- 6 $\lim_{x \rightarrow 2} \frac{3x^2 - x - 10}{x^2 - x - 2}$
- 7 $\lim_{x \rightarrow 2} \frac{x^4 - 16}{x^2 - x - 2}$
- 8 $\lim_{x \rightarrow 3^+} \frac{1}{x - 3}$
- 9 $\lim_{x \rightarrow 0^+} \frac{1}{\sqrt{x}}$
- 10 $\lim_{x \rightarrow 5} \frac{(1/x) - (1/5)}{x - 5}$
- 11 $\lim_{x \rightarrow 1/2} \frac{8x^3 - 1}{2x - 1}$
- 12 $\lim_{x \rightarrow 2} 5$
- 13 $\lim_{x \rightarrow 3^+} \frac{3 - x}{|3 - x|}$
- 14 $\lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2}$
- 15 $\lim_{h \rightarrow 0} \frac{(a + h)^4 - a^4}{h}$
- 16 $\lim_{x \rightarrow -3} \sqrt[3]{\frac{x + 3}{x^3 + 27}}$
- 17 $\lim_{h \rightarrow 0} \frac{(2 + h)^{-3} - 2^{-3}}{h}$
- 18 $\lim_{x \rightarrow 5} (x^2 + 3)^0$
- 19 $\lim_{x \rightarrow 0} \frac{x^2}{\sin x}$
- 20 $\lim_{x \rightarrow 0} \frac{x^2 + \sin^2 x}{4x^2}$
- 21 $\lim_{x \rightarrow 0} \frac{\sin^2 x + \sin 2x}{3x}$
- 22 $\lim_{x \rightarrow 0} \frac{2 - \cos x}{1 + \sin x}$
- 23 $\lim_{x \rightarrow 0} \frac{2 \cos x + 3x - 2}{5x}$
- 24 $\lim_{x \rightarrow 0} \frac{3x + 1 - \cos^2 x}{\sin x}$
- 25 $\lim_{x \rightarrow 0} \frac{x \sin x}{1 - \cos x}$
- 26 $\lim_{x \rightarrow 0} \frac{\cos x - 1}{2x}$

Find the limits in Exercises 27 and 28, where $[\]$ denotes the greatest integer function.

- 27 $\lim_{x \rightarrow 3^+} ([x] - x^2)$
- 28 $\lim_{x \rightarrow 3^-} ([x] - x^2)$
- 29 Prove, directly from the definition of limit, that $\lim_{x \rightarrow 6} (5x - 21) = 9$.
- 30 Suppose $f(x) = 1$ if x is rational and $f(x) = -1$ if x is irrational. Prove that $\lim_{x \rightarrow a} f(x)$ does not exist for any real number a .

In Exercises 31–34, find all numbers for which f is continuous.

- 31 $f(x) = 2x^4 - \sqrt[3]{x} + 1$
- 32 $f(x) = \sqrt{(2 + x)(3 - x)}$
- 33 $f(x) = \frac{\sqrt{9 - x^2}}{x^4 - 16}$
- 34 $f(x) = \frac{\sqrt{x}}{x^2 - 1}$

In Exercises 35–38 find the discontinuities of f .

- 35 $f(x) = \frac{|x^2 - 16|}{x^2 - 16}$
- 36 $f(x) = \frac{1}{x^2 - 16}$
- 37 $f(x) = \frac{x^2 - x - 2}{x^2 - 2x}$
- 38 $f(x) = \frac{x + 2}{x^3 - 8}$

In Exercises 39 and 40, prove that the function f is continuous at 0.

- 39 $f(x) = \frac{4 + \sin x}{x^2 + 2x + 5}$
- 40 $f(x) = (2x^2 + 3) \cos x$
- 41 If $f(x) = 1/x^2$, verify the Intermediate Value Theorem (2.29) for f on the interval $[2, 3]$.
- 42 Prove that the equation $x^5 + 7x^2 - 3x - 5 = 0$ has a root between -2 and -1 .

3 The Derivative

The *derivative of a function* is one of the most powerful tools in mathematics. Indeed, it is indispensable for nonelementary investigations in both the natural and human sciences. We shall begin our work by reformulating the notion of tangent line introduced in the pre-

ceding chapter and then discussing the velocity of a moving object. These two concepts serve to motivate the definition of derivative given in Section 3.2. The remainder of the chapter is concerned primarily with properties of the derivative.

3.1 Introduction

Let $P(a, f(a))$ be any point on the graph of a function f . Another point on the graph may be denoted by $Q(a + h, f(a + h))$, where h is the difference between the x -coordinates of Q and P (see (i) of Figure 3.1). By Definition (1.14), the slope m_{PQ} of the secant line through P and Q is

$$m_{PQ} = \frac{f(a + h) - f(a)}{h}.$$

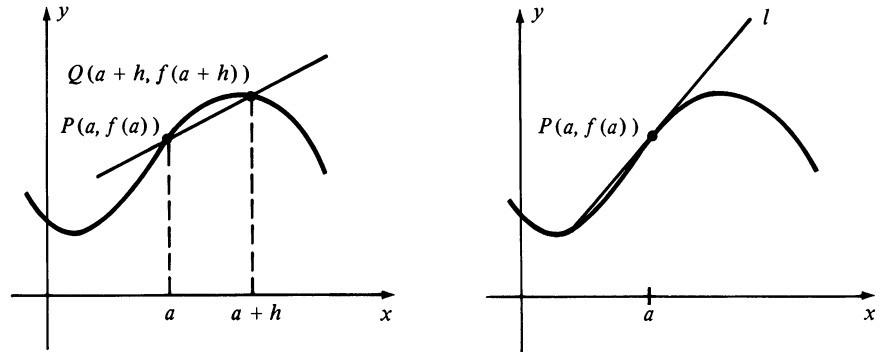
In Chapter 2 the slope m of the tangent line l at P was introduced as the limiting value of m_{PQ} as Q approaches P (see (2.2) and (ii) of Figure 3.1). If f is continuous, then we can make Q approach P by letting h approach 0. It is natural, therefore, to define m as follows.

Definition (3.1)

Let f be a function that is defined on an open interval containing a . The **slope m of the tangent line** to the graph of f at the point $P(a, f(a))$ is given by

$$m = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

provided the limit exists.



(i) Slope of secant line:

$$m_{PQ} = \frac{f(a+h) - f(a)}{h}$$

(ii) Slope of tangent line l :

$$m = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

FIGURE 3.1

If a tangent line is vertical, its slope is undefined and the limit in Definition (3.1) does not exist. Vertical tangent lines will be studied in the next chapter.

Example 1 If $f(x) = x^2$, find the slope of the tangent line to the graph of f at the point $P(a, a^2)$.

Solution This problem is the same as that stated for equations in Example 1 of Section 2.1 (see Figure 2.3). Using the quotient in Definition (3.1),

$$\begin{aligned} \frac{f(a+h) - f(a)}{h} &= \frac{(a+h)^2 - a^2}{h} \\ &= \frac{a^2 + 2ah + h^2 - a^2}{h} \\ &= \frac{2ah + h^2}{h} \\ &= 2a + h. \end{aligned}$$

The slope m of the tangent line is, therefore,

$$m = \lim_{h \rightarrow 0} (2a + h) = 2a. \quad \blacksquare$$

One of the main reasons for the invention of calculus was the need for a way to study objects in motion. Let us consider the problem of arriving at a satisfactory definition for the velocity, or speed, of an object at a given instant. We shall assume, for simplicity, that the object is moving on a line. Motion on a line is called **rectilinear motion**. It is easy to define the **average velocity** r during an interval of time. We merely use the following formula:

$$(3.2) \quad \text{Average velocity} = r = \frac{d}{t}$$

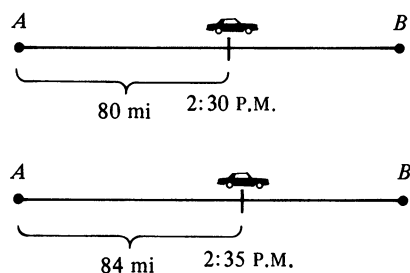


FIGURE 3.2

where t denotes the length of the time interval and d is the distance between the initial position of the object and its position after t units of time.

As an elementary illustration, suppose an automobile leaves city A at 1:00 P.M. and travels along a straight highway, arriving at city B , 150 miles from A , at 4:00 P.M. Employing (3.2) we see that its average velocity r during the indicated time interval is $150/3$, or 50 miles per hour. This is the velocity that, if maintained for three hours, would enable the automobile to travel the distance from A to B . The average velocity gives no information whatsoever about the velocity at any instant. For example, at 2:30 P.M. the automobile's speedometer may have registered 40, or 30, or the automobile may not even have been moving. If we wish to determine the rate at which the automobile is traveling at 2:30 P.M., information is needed about its motion or position *near* this time. For example, suppose at 2:30 P.M. the automobile is 80 miles from A and at 2:35 P.M. it is 84 miles from A , as illustrated in Figure 3.2. For the interval from 2:30 P.M. to 2:35 P.M. the elapsed time t is 5 minutes, or $1/12$ hour, and the distance d is 4 miles. Substituting in (3.2), the average velocity r during this time interval is

$$r = \frac{4}{(1/12)} = 48 \text{ miles per hour.}$$

However, this is still not an accurate indication of the velocity at 2:30 P.M. since, for example, the automobile may have been traveling very slowly at 2:30 P.M. and then speeded up considerably so as to arrive at the point 84 miles from A at 2:35 P.M. Evidently, a better approximation of the motion would be obtained by considering the average velocity during a smaller time interval, say from 2:30 P.M. to 2:31 P.M. Indeed, it appears that the best procedure would be to take smaller and smaller time intervals near 2:30 P.M. and study the average velocity in each time interval. This leads us into a limiting process similar to that discussed for tangent lines.

In order to base our discussion on mathematical concepts, let us assume that the position of an object moving rectilinearly may be represented by a point P on a coordinate line l . We shall sometimes refer to the motion of the point P on l , or the motion of a *particle* on l whose position is specified by P . We further assume that the position of P is known at every instant in a given interval of time. If $f(t)$ denotes the coordinate of P at time t , then the function f determined in this way is called the **position function** for P . If we keep track of time by means of a clock, then for each t the point P is $f(t)$ units from the origin, as illustrated in Figure 3.3.

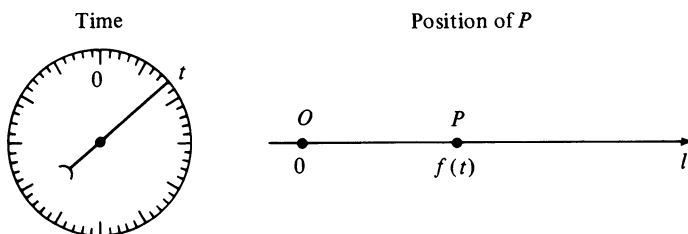


FIGURE 3.3

To define the velocity of P at time a , we begin by investigating the average velocity in a (small) time interval near a . Thus we consider times a and $a + h$ where h may be positive or negative but not zero. The corresponding positions of P on the l -axis are given by $f(a + h)$ and $f(a)$, as illustrated in Figure 3.4, and hence the amount of change in the position of P is $f(a + h) - f(a)$. The latter number may be positive, negative, or zero, depending on whether the position of P at time $a + h$ is to the right, to the left, or the same as its position at time a . This number is not necessarily the distance traversed by P during the time interval $[a, a + h]$ since, for example, P may have moved beyond the point corresponding to $f(a + h)$ and then returned to that point at time $a + h$.

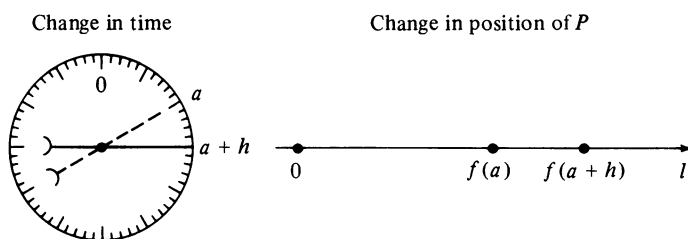


FIGURE 3.4

By (3.2), the average velocity of P during the time interval $[a, a + h]$ is given by

$$\text{Average velocity} = \frac{f(a + h) - f(a)}{h}.$$

As in our previous discussion, the smaller h is numerically, the closer this quotient should approximate the velocity of P at time a . Accordingly, we *define* the velocity as the limit, as h approaches zero, of the average velocity, provided the limit exists. This limit is also called the *instantaneous velocity* of P at time a . Summarizing, we have the following definition.

Definition (3.3)

If a point P moves on a coordinate line l such that its coordinate at time t is $f(t)$, then the **velocity** $v(a)$ of P at time a is given by

$$v(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

provided the limit exists.

If $f(t)$ is measured in centimeters and t in seconds, then the unit of velocity is centimeters per second, abbreviated cm/sec. If $f(t)$ is in miles and t in hours, then velocity is in miles per hour (mi/hr). Other units of measurement may, of course, be used.

We shall return to the velocity concept in Chapter 4, where it will be shown that if the velocity is positive in a given time interval, then the point is moving in the positive direction on l , whereas if the velocity is negative, the point is moving in the negative direction. Although these facts have not been proved we shall use them in the following example.

Example 2 The position of a point P on a coordinate line l is given by $f(t) = t^2 - 6t$, where $f(t)$ is measured in feet and t in seconds. Find the velocity at time a . What is the velocity at $t = 0$? At $t = 4$? Determine time intervals in which (a) P moves in the positive direction on l and (b) P moves in the negative direction. At what time is the velocity 0?

Solution From the formula for $f(t)$ we obtain

$$\begin{aligned}\frac{f(a+h) - f(a)}{h} &= \frac{[(a+h)^2 - 6(a+h)] - [a^2 - 6a]}{h} \\ &= \frac{a^2 + 2ah + h^2 - 6a - 6h - a^2 + 6a}{h} \\ &= \frac{2ah + h^2 - 6h}{h} \\ &= 2a + h - 6.\end{aligned}$$

Consequently, by Definition (3.3), the velocity $v(a)$ at time a is

$$v(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} (2a + h - 6) = 2a - 6.$$

In particular, the velocity at $t = 0$ is $v(0) = 2(0) - 6 = -6$ ft/sec. At $t = 4$ it is $v(4) = 2(4) - 6 = 2$ ft/sec.

According to the remarks preceding this example, P moves to the left when the velocity is negative; that is, when $2a - 6 < 0$, or $a < 3$. The particle moves to the right when $2a - 6 > 0$ or $a > 3$. In terms of intervals, the motion is to the left in the time interval $(-\infty, 3)$ and to the right in $(3, \infty)$. The velocity is zero when $2a - 6 = 0$, that is, when $a = 3$ seconds. ■

You have undoubtedly noted the similarity of the limit in Definition (3.3) to that used in the definition of tangent line in (3.1). Indeed, the two expressions are identical! There are many different mathematical and physical applications which lead to precisely this same limit. Several of these will be discussed in the next chapter.

3.1 Exercises

In Exercises 1–4 find the slope of the tangent line at the point $P(a, f(a))$ on the graph of f . Sketch the graph and show the tangent lines at various points.

1 $f(x) = 2 - x^3$

2 $f(x) = 3x - 5$

3 $f(x) = \sqrt{x} + 1$

4 $f(x) = (1/x) - 1$

In Exercises 5 and 6, the position of a point P moving on a coordinate line l is given by $f(t)$, where t is measured in seconds and $f(t)$ in centimeters.

- (a) Find the average velocity of P in the following time intervals: $[1, 1.2]$; $[1, 1.1]$; $[1, 1.01]$; $[1, 1.001]$.
- (b) Find the velocity of P at $t = 1$.
- (c) Determine the time intervals in which P moves in the positive direction.
- (d) Determine the time intervals in which P moves in the negative direction.

5 $f(t) = 4t^2 + 3t$

6 $f(t) = t^3$

- 7 A projectile is fired directly upward from the ground with an initial velocity of 112 ft/sec, and its distance above the ground after t seconds is $112t - 16t^2$ ft. What

is the velocity of the projectile at $t = 2$, $t = 3$, and $t = 4$? At what time does it reach its maximum height? When does it strike the ground? What is its velocity at the moment of impact?

- 8 If a balloonist drops ballast (a sand bag) from a balloon 500 ft above the ground, then its distance above ground after t seconds is $500 - 16t^2$ ft. Find the velocity at $t = 1$, $t = 2$, and $t = 3$. With what velocity does the ballast strike the ground?
- 9 If the position of an object moving on a coordinate line is given by a polynomial function of degree 1, prove that the velocity is constant.
- 10 If the position function of a rectilinearly moving object is a constant function, prove that the velocity is 0 at all times. Describe the motion of the particle.

3.2 The Derivative of a Function

In the preceding section the same limiting process occurred in two different situations. The limit in Definitions (3.1) and (3.3) is one of the fundamental concepts of calculus. In the remainder of the chapter a number of rules pertaining to this concept will be developed. Let us begin by introducing the terminology and notation associated with this limit.

Definition (3.4)

Let f be a function that is defined on an open interval containing a . The **derivative of f at a** , written $f'(a)$, is given by

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

provided the limit exists.

The symbol $f'(a)$ is read “ f prime of a .” The terminology “ $f'(a)$ exists” will mean that the limit in Definition (3.4) exists. If $f'(a)$ exists we say that the function f is **differentiable at a** or **f has a derivative at a** .

It is important to observe that if f is differentiable at a , then by Definition (3.1), $f'(a)$ is the slope of the tangent line to the graph of f at the point $(a, f(a))$.

We say that a function f is **differentiable on an open interval (a, b)** if it is differentiable at every number c in (a, b) . In like manner, we refer to functions that are differentiable on intervals of the form (a, ∞) , $(-\infty, a)$, or $(-\infty, \infty)$. For closed intervals we use the following convention, which is analogous to the definition of continuity on a closed interval given in (2.25).

Definition (3.5)

A function f is **differentiable on a closed interval** $[a, b]$ if it is differentiable on (a, b) and if the following limits exist:

$$\lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h} \quad \text{and} \quad \lim_{h \rightarrow 0^-} \frac{f(b+h) - f(b)}{h}.$$

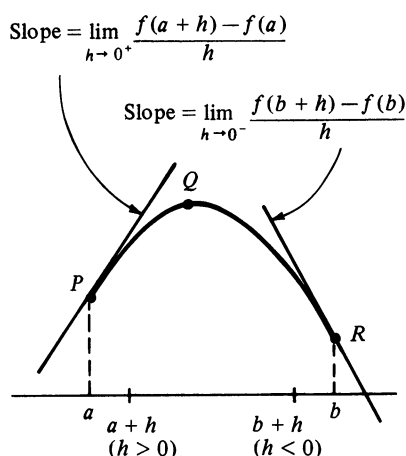


FIGURE 3.5

The one-sided limits specified in Definition (3.5) are sometimes referred to as the **right-hand** and **left-hand derivatives** of f at a and b , respectively. Note that for the right-hand derivative we have $h \rightarrow 0^+$. Consequently $h > 0$ and $a + h > a$. Thus we may think of $a + h$ as approaching a from the right. For the left-hand derivative, $h \rightarrow 0^-$ and in this case, $h < 0$ and $b + h < b$. Hence for the left-hand derivative we may regard $b + h$ as approaching b from the left.

If f is defined on a closed interval $[a, b]$ and is undefined outside of this interval, then the right-hand and left-hand derivatives allow us to define the slopes of the tangent lines at the points $P(a, f(a))$ and $R(b, f(b))$, respectively, as illustrated in Figure 3.5. Thus, for the slope of the tangent line at P we take the limiting value of the slope m_{PQ} of the secant line through P and Q as Q approaches P from the right. For the tangent line at R , the point Q approaches R from the left.

Differentiability on an interval of the form $[a, b)$, $[a, \infty)$, $(a, b]$, or $(-\infty, b]$ is defined in the obvious way, using a one-sided limit at an endpoint.

It should be clear that if f is defined on an open interval containing a , then $f'(a)$ exists if and only if both the right-hand and left-hand derivatives exist at a , and are equal. The functions whose graphs are sketched in Figure 3.6 have right-hand and left-hand derivatives at a which give the slopes of the indicated lines l_1 and l_2 , respectively. However, since the slopes of l_1 and l_2 are unequal, $f'(a)$ does not exist. Generally, if the graph of f has a corner at $(a, f(a))$, then f is not differentiable at a .

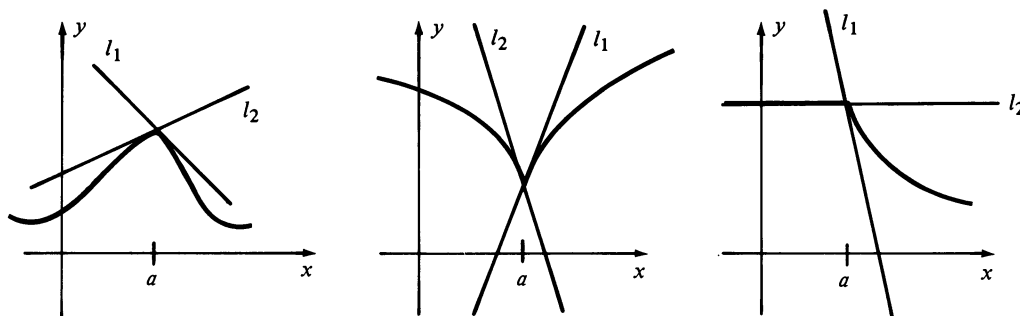


FIGURE 3.6

Given a function f , let $S = \{a: f'(a) \text{ exists}\}$. If f is defined only on certain intervals, then the terminology " $f'(a)$ exists" may refer to the existence of certain right-hand or left-hand limits of the type specified in Definition (3.5). If, with each x in S , we associate the number $f'(x)$, we obtain a function f'

with domain S called the **derivative of f** . The value of f' at x is given by the following (or an appropriate one-sided) limit:

$$(3.6) \quad f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

It is important to note that in determining $f'(x)$, the number x is fixed, but arbitrary, and the limit is taken as h approaches zero. The terminology “differentiate $f(x)$ ” is used interchangeably with “find the derivative of $f(x)$.” In statements of definitions and theorems, the expression “ f is differentiable” will mean that $f'(x)$ exists for all numbers x in some set of real numbers. The domain of f' will not always be stated explicitly; however, for specific problems, it may often be found by inspection.

Example 1 If $f(x) = 3x^2 - 5x + 4$, find $f'(x)$. What is the domain of f' ? Find $f'(2)$, $f'(-\sqrt{2})$, and $f'(a)$.

Solution Employing (3.6),

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[3(x+h)^2 - 5(x+h) + 4] - (3x^2 - 5x + 4)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(3x^2 + 6xh + 3h^2 - 5x - 5h + 4) - (3x^2 - 5x + 4)}{h} \\ &= \lim_{h \rightarrow 0} \frac{6xh + 3h^2 - 5h}{h} = \lim_{h \rightarrow 0} (6x + 3h - 5) \\ &= 6x - 5. \end{aligned}$$

Since $f'(x) = 6x - 5$ for all x , the domain of f' is $\mathbb{R}$. Substituting for x , we have

$$\begin{aligned} f'(2) &= 6(2) - 5 = 7 \\ f'(-\sqrt{2}) &= 6(-\sqrt{2}) - 5 = -(6\sqrt{2} + 5) \\ f'(a) &= 6a - 5. \end{aligned}$$

■

Example 2 Find $f'(x)$ if $f(x) = \sqrt{x}$. What is the domain of f' ?

Solution The domain of f consists of all nonnegative real numbers. We shall examine the cases $x > 0$ and $x = 0$ separately. If $x > 0$, then by (3.6),

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}.$$

To find the limit we begin by multiplying numerator and denominator of the indicated quotient by $\sqrt{x+h} + \sqrt{x}$. Thus

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} \\ &= \lim_{h \rightarrow 0} \frac{(x+h) - x}{h(\sqrt{x+h} + \sqrt{x})} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} \\ &= \frac{1}{\sqrt{x} + \sqrt{x}} = \frac{1}{2\sqrt{x}}. \end{aligned}$$

Since $x = 0$ is an endpoint of the domain of f , we must use a one-sided limit to determine whether $f'(0)$ exists. Specifically, if f is differentiable at 0, then using Definition (3.5) with $x = 0$,

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0^+} \frac{\sqrt{0+h} - \sqrt{0}}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{\sqrt{h}}{h} = \lim_{h \rightarrow 0^+} \frac{1}{\sqrt{h}}. \end{aligned}$$

Since the last limit does not exist (see Exercise 38 of Section 2.3), $f'(0)$ does not exist. Hence the domain of f' is the set of positive real numbers. ■

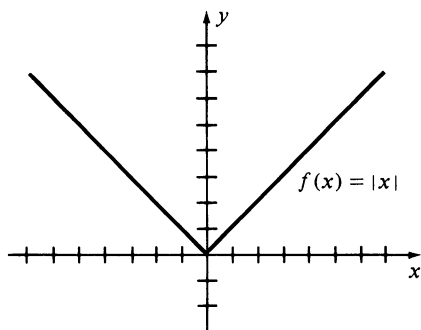


FIGURE 3.7

Example 3 Show that the function f defined by $f(x) = |x|$ is not differentiable at 0.

Solution The graph of f was discussed in Example 5 of Section 1.4 and is resketched in Figure 3.7.

It is geometrically evident that f has no derivative at 0 since the graph has a corner at the origin. We can prove that $f'(0)$ does not exist by showing that the right-hand and left-hand derivatives of f at 0 are not equal. Using the limits in Definition (3.5) with $a = 0$ and $b = 0$,

$$\lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{|0+h| - |0|}{h} = \lim_{h \rightarrow 0^+} \frac{|h|}{h} = 1$$

$$\lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{|0+h| - |0|}{h} = \lim_{h \rightarrow 0^-} \frac{|h|}{h} = -1.$$

Thus $f'(0)$ does not exist. ■

It follows from Example 3 that the graph of $y = |x|$ does not have a tangent line at the point $P(0, 0)$.

In Definition (3.4), the derivative was defined as a certain limit. There is another important limit formula for $f'(a)$. To see how it arises geometrically, let us begin by labeling the graph of f as shown in Figure 3.8.

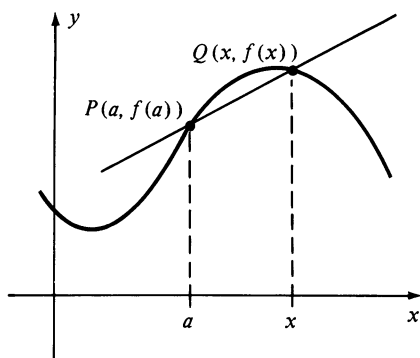


FIGURE 3.8

By Definition (1.14), the slope m_{PQ} of the secant line through P and Q is

$$m_{PQ} = \frac{f(x) - f(a)}{x - a}.$$

If f is continuous at a , we can make Q approach P by letting x approach a . Thus it appears that the slope of the tangent line at P is given by

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

provided the limit exists. Since $m = f'(a)$ this leads to the alternative formula for the derivative of f stated in the next theorem. A nongeometric proof of the formula may be found in Appendix II.

Theorem (3.7)

If f is defined on an open interval containing a , then

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

provided the limit exists.

Example 4 If $f(x) = x^{1/3}$ and $a \neq 0$, find $f'(a)$.

Solution By Theorem (3.7),

$$f'(a) = \lim_{x \rightarrow a} \frac{x^{1/3} - a^{1/3}}{x - a}$$

provided the limit exists. We now change the form of the indicated quotient as follows:

$$\lim_{x \rightarrow a} \frac{x^{1/3} - a^{1/3}}{x - a} = \lim_{x \rightarrow a} \frac{x^{1/3} - a^{1/3}}{(x^{1/3})^3 - (a^{1/3})^3}.$$

We may factor the denominator by using the formula

$$p^3 - q^3 = (p - q)(p^2 + pq + q^2)$$

with $p = x^{1/3}$ and $q = a^{1/3}$. This gives us

$$\lim_{x \rightarrow a} \frac{x^{1/3} - a^{1/3}}{(x^{1/3} - a^{1/3})(x^{2/3} + x^{1/3}a^{1/3} + a^{2/3})}.$$

Dividing numerator and denominator of the last quotient by $x^{1/3} - a^{1/3}$ and taking the limit, we obtain

$$f'(a) = \frac{1}{a^{2/3} + a^{2/3} + a^{2/3}} = \frac{1}{3a^{2/3}}.$$

■

Theorem (3.8)If a function f is differentiable at a , then f is continuous at a .

Proof If x is in the domain of f and $x \neq a$, then $f(x)$ may be written as follows:

$$f(x) = f(a) + \frac{f(x) - f(a)}{x - a}(x - a).$$

Employing limit theorems and Theorem (3.7),

$$\begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} f(a) + \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot \lim_{x \rightarrow a} (x - a) \\ &= f(a) + f'(a) \cdot 0 = f(a). \end{aligned}$$

Thus, by Definition (2.23), f is continuous at a . □

The converse of Theorem (3.8) is false; that is, *there exist continuous functions that are not differentiable*. The function in Example 4 is continuous but not differentiable at 0. Similarly, if $f(x) = |x|$, then f is continuous at 0; however, as was proved in Example 3, f is not differentiable at 0.

We shall conclude this section by introducing the following notations for derivatives, where $y = f(x)$:

$$(3.9) \quad f'(x) = D_x[f(x)] = D_x y = y'.$$

In (3.9) the subscript x on D is employed to designate the independent variable. For example, if the independent variable is t we write $f'(t) = D_t[f(t)]$. The symbols D_x , D_t , etc., are referred to as **differential operators**. Standing alone, D_x has no practical significance; however, if an expression in x is placed to the right of it, then the derivative is obtained. To illustrate, using Example 1,

$$D_x(3x^2 - 5x + 4) = 6x - 5.$$

We say that D_x *operates* on the expression $3x^2 - 5x + 4$. Sometimes $D_x y$ is referred to as **the derivative of y with respect to x** . As indicated in (3.9), y' may be used as an abbreviation for this derivative.

In Section 3.5 we shall introduce still another notation for derivatives. All of these representations are used in mathematics and applications, and it is advisable for students to become familiar with the various forms.

3.2 Exercises

In Exercises 1–10 use (3.6) to find $f'(x)$.

1 $f(x) = 37$

2 $f(x) = 17 - 6x$

5 $f(x) = 2 + 8x - 5x^2$

6 $f(x) = x^3 + x$

3 $f(x) = 9x - 2$

4 $f(x) = 7x^2 - 5$

7 $f(x) = 1/(x - 2)$

8 $f(x) = (1 + \sqrt{3})^2$

9 $f(x) = \sqrt{3x + 1}$

10 $f(x) = 1/2x$

In Exercises 11–14 find $D_x y$.

11 $y = 7/\sqrt{x}$

12 $y = (2x + 3)^2$

13 $y = 2x^3 - 4x + 1$

14 $y = x/(3x + 4)$

In Exercises 15–20 find $f'(a)$ by means of Theorem (3.7).

15 $f(x) = x^2$

16 $f(x) = \sqrt{2}x$

17 $f(x) = 6/x^2$

18 $f(x) = 8 - x^3$

19 $f(x) = 1/(x + 5)$

20 $f(x) = \sqrt{x}$

In Exercises 21 and 22 use right-hand and left-hand derivatives to prove that f is not differentiable at $x = 5$.

21 $f(x) = |x - 5|$

22 $f(x) = [x]$ (the greatest integer function)

23 Prove that if n is any integer, then the greatest integer function f is not differentiable at n .

24 Suppose $f(x) = 0$ if x is rational and $f(x) = 1$ if x is irrational. Prove that f is not differentiable at any real number a .

25 If $f(x)$ is a polynomial of degree 1, prove that $f'(x)$ is a polynomial of degree 0. What if $f(x)$ is a polynomial of degree 2 or 3?

26 Prove that for every real number c , $D_x c = 0$.

In Exercises 27–30, sketch the graph of f and find the domain of f' .

27 $f(x) = \begin{cases} 2x & \text{if } x \leq 0 \\ x^2 & \text{if } x > 0 \end{cases}$

28 $f(x) = \begin{cases} 2x - 1 & \text{if } x \leq 1 \\ x^2 & \text{if } x > 1 \end{cases}$

29 $f(x) = \begin{cases} |x| & \text{if } |x| \leq 1 \\ 2 - |x| & \text{if } |x| > 1 \end{cases}$

30 $f(x) = \begin{cases} x - [x] & \text{if } n \leq x < n + 1 \text{ and } n \text{ is an even integer} \\ 1 - x + [x] & \text{if } n \leq x < n + 1 \text{ and } n \text{ is an odd integer} \end{cases}$

where $[]$ denotes the greatest integer function.

31 Let $f(x) = |x|$. Prove that $f'(x) = 1$ if $x > 0$ and that $f'(x) = -1$ if $x < 0$.

32 Let $f(x) = |x|/x$. Find (a) the domain of f' ; (b) $f'(x)$ for every x in the domain of f' .

3.3 Rules for Finding Derivatives

This section contains some general rules that simplify the task of finding derivatives. In statements of theorems we shall use the differential operator symbol D_x to denote derivatives (see (3.9)). The first result is sometimes phrased: *the derivative of a constant is zero*.

Theorem (3.10)

$$D_x c = 0$$

Proof Let f be the constant function such that $f(x) = c$ for every x . We wish to show that $f'(x) = 0$. Since every value of f is c it follows that $f(x + h) = c$ for all h . Hence

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} 0 = 0$$

where the last step follows from Theorem (2.5). □

Theorem (3.11)

$$D_x(x) = 1$$

Proof If we let $f(x) = x$, then using (3.6),

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h) - x}{h} \\ &= \lim_{h \rightarrow 0} \frac{h}{h} = \lim_{h \rightarrow 0} 1 = 1. \end{aligned}$$

□

In the proof of the next rule we shall make use of the *Binomial Theorem*, which states that the following formula is true for all real numbers a and b and every positive integer n .

$$\begin{aligned} (3.12) \quad (a+b)^n &= a^n + na^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 \\ &\quad + \cdots + \binom{n}{r}a^{n-r}b^r + \cdots + nab^{n-1} + b^n \end{aligned}$$

where the symbol $\binom{n}{r}$ used for the coefficient of the term involving b^r is defined by

$$\binom{n}{r} = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r(r-1)(r-2)\cdots 1} = \frac{n!}{(n-r)!r!}.$$

The validity of (3.12) may be established by mathematical induction. The special cases $n = 2$, $n = 3$, and $n = 4$ are

$$\begin{aligned} (a+b)^2 &= a^2 + 2ab + b^2 \\ (a+b)^3 &= a^3 + 3a^2b + 3ab^2 + b^3 \\ (a+b)^4 &= a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4. \end{aligned}$$

The Power Rule (3.13)

$$\text{If } n \text{ is a positive integer, then } D_x(x^n) = nx^{n-1}.$$

Proof Let $f(x) = x^n$. We wish to show that $f'(x) = nx^{n-1}$. By (3.6),

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}$$

provided the limit exists. Employing the Binomial Theorem (3.12) with $a = x$ and $b = h$ gives us

$$(x + h)^n = x^n + nx^{n-1}h + \frac{n(n-1)}{2!}x^{n-2}h^2 + \cdots + nxh^{n-1} + h^n.$$

It is important to observe that every term after the first contains h to some positive integral power. If we subtract x^n and divide by h we obtain

$$f'(x) = \lim_{h \rightarrow 0} \left[nx^{n-1} + \frac{n(n-1)}{2!}x^{n-2}h + \cdots + nxh^{n-2} + h^{n-1} \right].$$

Since each term within the brackets except the first contains a power of h , we see that $f'(x) = nx^{n-1}$.

If $x \neq 0$, then (3.13) is also true if $n = 0$, for in this case $f(x) = x^0 = 1$ and by Theorem (3.10), $f'(x) = 0 = 0 \cdot x^{0-1}$. $\square$

Example 1

(a) Find $D_x(x^3)$ and $D_x(x^8)$.

(b) Find y' if $y = x^{100}$.

Solution Applying the Power Rule (3.13) gives us

(a) $D_x(x^3) = 3x^2$ and $D_x(x^8) = 8x^7$.

(b) $y' = D_x y = D_x(x^{100}) = 100x^{99}$. $\blacksquare$

If symbols other than x are used for the independent variable, then the Power Rule may be written as

$$D_t(t^n) = nt^{n-1}, \quad D_z(z^n) = nz^{n-1}, \quad D_v(v^n) = nv^{n-1},$$

etc. In Section 3.7 we shall prove that the Power Rule is valid for every rational number n .

In the statements of Theorems (3.14)–(3.17) it is assumed that the functions f and g are differentiable.

Theorem (3.14)

$$D_x[cf(x)] = cD_x[f(x)]$$

Proof If we let $g(x) = cf(x)$, then

$$\begin{aligned} D_x[cf(x)] &= D_x[g(x)] \\ &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{cf(x+h) - cf(x)}{h} \\ &= c \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= cf'(x) = cD_x[f(x)]. \end{aligned}$$

$\square$

For the special case $f(x) = x^n$, Theorems (3.14) and (3.13) lead to the next formula, which is true for every real number c and every positive integer n :

$$D_x(cx^n) = cnx^{n-1}$$

Example 2

- (a) Find $D_x(7x^4)$.
 (b) Find $F'(z)$ if $F(z) = -3z^{15}$.

Solution By the remarks preceding this example we have

- (a) $D_x(7x^4) = (7)4x^3 = 28x^3$.
 (b) $F'(z) = (-3)(15)z^{14} = -45z^{14}$. ■

The theorems that follow involve several functions. As usual we assume that every number x is chosen from the intersection of the domains of the functions.

Theorem (3.15)

$$\begin{aligned} D_x[f(x) + g(x)] &= D_x[f(x)] + D_x[g(x)] \\ D_x[f(x) - g(x)] &= D_x[f(x)] - D_x[g(x)] \end{aligned}$$

Proof Let $k(x) = f(x) + g(x)$. We wish to show that $k'(x) = f'(x) + g'(x)$. This may be done as follows.

$$\begin{aligned} k'(x) &= \lim_{h \rightarrow 0} \frac{k(x+h) - k(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[f(x+h) + g(x+h)] - [f(x) + g(x)]}{h} \\ &= \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h} \right] \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\ &= f'(x) + g'(x) \end{aligned}$$

The formula for $D_x[f(x) - g(x)]$ is proved in similar fashion. □

The preceding theorem may be extended to sums or differences of any number of functions.

Since a polynomial is a sum of terms of the form cx^n where n is a non-negative integer, we may use results on sums and differences to obtain the derivative, as illustrated in the next example.

Example 3 Find $f'(x)$ if $f(x) = 2x^4 - 5x^3 + x^2 - 4x + 1$.

Solution

$$\begin{aligned} f'(x) &= D_x(2x^4 - 5x^3 + x^2 - 4x + 1) \\ &= D_x(2x^4) - D_x(5x^3) + D_x(x^2) - D_x(4x) + D_x(1) \\ &= 8x^3 - 15x^2 + 2x - 4 \end{aligned}$$

■

The Product Rule (3.16)

$$D_x[f(x)g(x)] = f(x)D_x[g(x)] + g(x)D_x[f(x)].$$

Proof Let $k(x) = f(x)g(x)$. We wish to show that

$$k'(x) = f(x)g'(x) + g(x)f'(x).$$

If $k'(x)$ exists, then

$$\begin{aligned} k'(x) &= \lim_{h \rightarrow 0} \frac{k(x+h) - k(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h}. \end{aligned}$$

To change the form of the quotient so that the limit may be evaluated, we subtract and add the expression $f(x+h)g(x)$ in the numerator. Thus

$$k'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h}$$

which may be written

$$\begin{aligned} k'(x) &= \lim_{h \rightarrow 0} \left[f(x+h) \cdot \frac{g(x+h) - g(x)}{h} + g(x) \cdot \frac{f(x+h) - f(x)}{h} \right] \\ &= \lim_{h \rightarrow 0} f(x+h) \cdot \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\ &\quad + \lim_{h \rightarrow 0} g(x) \cdot \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}. \end{aligned}$$

Since f is differentiable at x , it is continuous and hence $\lim_{h \rightarrow 0} f(x+h) = f(x)$. Also, $\lim_{h \rightarrow 0} g(x) = g(x)$ since x is fixed in this limiting process. Finally, applying the definition of derivative to $f(x)$ and $g(x)$ we obtain

$$k'(x) = f(x)g'(x) + g(x)f'(x).$$

□

The Product Rule may be phrased as follows: *The derivative of a product equals the first factor times the derivative of the second factor, plus the second times the derivative of the first.*

Example 4 Find $f'(x)$ if $f(x) = (x^3 + 1)(2x^2 + 8x - 5)$.

Solution Using the Product Rule (3.16),

$$\begin{aligned} f'(x) &= (x^3 + 1)D_x(2x^2 + 8x - 5) + (2x^2 + 8x - 5)D_x(x^3 + 1) \\ &= (x^3 + 1)(4x + 8) + (2x^2 + 8x - 5)(3x^2) \\ &= 4x^4 + 8x^3 + 4x + 8 + 6x^4 + 24x^3 - 15x^2 \\ &= 10x^4 + 32x^3 - 15x^2 + 4x + 8. \end{aligned}$$

We could also find $f'(x)$ by first multiplying the two factors $x^3 + 1$ and $2x^2 + 8x - 5$ and then differentiating the resulting polynomial. ■

The Quotient Rule (3.17)

$$D_x \left[\frac{f(x)}{g(x)} \right] = \frac{g(x)D_x[f(x)] - f(x)D_x[g(x)]}{[g(x)]^2},$$

provided $g(x) \neq 0$.

Proof Let $k(x) = f(x)/g(x)$. We wish to show that

$$k'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}.$$

Using the definitions of $k'(x)$ and $k(x)$,

$$\begin{aligned} k'(x) &= \lim_{h \rightarrow 0} \frac{k(x+h) - k(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} \\ &= \lim_{h \rightarrow 0} \frac{g(x)f(x+h) - f(x)g(x+h)}{hg(x+h)g(x)}. \end{aligned}$$

Subtracting and adding $g(x)f(x)$ in the numerator of the last quotient,

$$k'(x) = \lim_{h \rightarrow 0} \frac{g(x)f(x+h) - g(x)f(x) + g(x)f(x) - f(x)g(x+h)}{hg(x+h)g(x)}$$

or equivalently,

$$k'(x) = \lim_{h \rightarrow 0} \frac{g(x) \left[\frac{f(x+h) - f(x)}{h} \right] - f(x) \left[\frac{g(x+h) - g(x)}{h} \right]}{g(x+h)g(x)}.$$

Taking the limit of the numerator and denominator gives us the desired formula. □

The Quotient Rule may be stated as follows: *The derivative of a quotient equals the denominator times the derivative of the numerator minus the numerator times the derivative of the denominator, divided by the square of the denominator.*

Example 5 Find y' if $y = \frac{3x^2 - x + 2}{4x^2 + 5}$.

Solution By the Quotient Rule (3.17),

$$\begin{aligned} y' &= \frac{(4x^2 + 5)D_x(3x^2 - x + 2) - (3x^2 - x + 2)D_x(4x^2 + 5)}{(4x^2 + 5)^2} \\ &= \frac{(4x^2 + 5)(6x - 1) - (3x^2 - x + 2)(8x)}{(4x^2 + 5)^2} \\ &= \frac{(24x^3 - 4x^2 + 30x - 5) - (24x^3 - 8x^2 + 16x)}{(4x^2 + 5)^2} \\ &= \frac{4x^2 + 14x - 5}{(4x^2 + 5)^2}. \end{aligned}$$

It is now a simple matter to extend the Power Rule (3.13) to the case in which the exponent is a negative integer.

Theorem (3.18)

If n is a positive integer, then $D_x(x^{-n}) = -nx^{-n-1}$.

Proof Using the definition of x^{-n} and the Quotient Rule (3.17),

$$\begin{aligned} D_x(x^{-n}) &= D_x\left(\frac{1}{x^n}\right) = \frac{x^n D_x(1) - 1 D_x(x^n)}{(x^n)^2} \\ &= \frac{x^n(0) - 1(nx^{n-1})}{(x^n)^2} = \frac{-nx^{n-1}}{x^{2n}} \\ &= (-n)x^{(n-1)-2n} = (-n)x^{-n-1}. \end{aligned}$$

Example 6 Differentiate the following: (a) $g(w) = 1/w^4$. (b) $H(s) = 3/s$.

Solution

(a) Writing $g(w) = w^{-4}$ and using Theorem (3.18) (with w as the independent variable),

$$g'(w) = D_w(w^{-4}) = -4w^{-5} = -\frac{4}{w^5}.$$

(b) Since $H(s) = 3s^{-1}$,

$$H'(s) = D_s(3s^{-1}) = 3(-1)s^{-2} = -\frac{3}{s^2}.$$

The differentiation formulas (3.15), (3.16) and (3.17) are stated in terms of arbitrary values $f(x)$ and $g(x)$ of the functions f and g . If we wish to state these rules without referring to the variable x , we may use the following forms, where it is assumed that f and g are differentiable functions.

$$(3.19) \quad \begin{aligned} (f + g)' &= f' + g' & (f - g)' &= f' - g' \\ (fg)' &= fg' + gf' & \left(\frac{f}{g}\right)' &= \frac{gf' - fg'}{g^2} \end{aligned}$$

3.3 Exercises

Differentiate the functions defined in Exercises 1–32.

- 1 $f(x) = 10x^2 + 9x - 4$
- 2 $f(x) = 6x^3 - 5x^2 + x + 9$
- 3 $f(s) = 15 - s + 4s^2 - 5s^4$
- 4 $f(t) = 12 - 3t^4 + 4t^6$
- 5 $g(x) = (x^3 - 7)(2x^2 + 3)$
- 6 $k(x) = (2x^2 - 4x + 1)(6x - 5)$
- 7 $h(r) = r^2(3r^4 - 7r + 2)$
- 8 $g(s) = (s^3 - 5s + 9)(2s + 1)$
- 9 $f(x) = (4x - 5)/(3x + 2)$
- 10 $h(x) = (8x^2 - 6x + 11)/(x - 1)$
- 11 $h(z) = (8 - z + 3z^2)/(2 - 9z)$
- 12 $f(w) = 2w/(w^3 - 7)$
- 13 $f(x) = 3x^3 - 2x^2 + 4x - 7$
- 14 $g(z) = 5z^4 - 8z^2 + z$
- 15 $F(t) = t^2 + (1/t^2)$
- 16 $s(x) = 2x + 1/(2x)$
- 17 $g(x) = (8x^2 - 5x)(13x^2 + 4)$
- 18 $H(y) = (y^5 - 2y^3)(7y^2 + y - 8)$
- 19 $G(v) = (v^3 - 1)/(v^3 + 1)$
- 20 $f(t) = (8t + 15)/(t^2 - 2t + 3)$
- 21 $f(x) = 1/(1 + x + x^2 + x^3)$
- 22 $p(x) = 1 + (1/x) + (1/x^2) + (1/x^3)$
- 23 $g(z) = z(2z^3 - 5z - 1)(6z^2 + 7)$

- 24 $N(v) = 4v(v - 1)(2v - 3)$
- 25 $K(s) = (3s)^{-4}$
- 26 $W(s) = (3s)^4$
- 27 $h(x) = (5x - 4)^2$
- 28 $g(r) = (5r - 4)^{-2}$
- 29 $f(t) = \frac{(3/5t) - 1}{(2/t^2) + 7}$
- 30 $S(w) = (2w + 1)^3$
- 31 $M(x) = (2x^3 - 7x^2 + 4x + 3)/x^2$
- 32 $f(x) = (3x^2 - 5x + 8)/7$

In Exercises 33 and 34 find y' by means of (a) the Quotient Rule (3.17), (b) the Product Rule (3.16), and (c) simplifying algebraically and using the Power Rule (3.13).

- 33 $y = (3x - 1)/x^2$
- 34 $y = (x^2 + 1)/x^4$

In Exercises 35 and 36 find y' by (a) using the Product Rule and (b) first multiplying the two factors.

- 35 $y = (12x - 17)(5x + 3)$
- 36 $y = 8x^2(5x - 9)$

37 If f , g , and h are differentiable, use the Product Rule to prove that

$$D_x[f(x)g(x)h(x)] = f(x)g(x)h'(x) + f(x)h(x)g'(x) + h(x)g(x)f'(x).$$

As a corollary, let $f = g = h$ to prove that

$$D_x[f(x)]^3 = 3[f(x)]^2 f'(x).$$

- 38 Extend Exercise 37 to the derivative of a product of four functions and then find a formula for $D_x[f(x)]^4$.

In Exercises 39 and 40, use Exercise 37 to find y' .

- 39 $y = (8x - 1)(x^2 + 4x + 7)(x^3 - 5)$
- 40 $y = (3x^4 - 10x^2 + 8)(2x^2 - 10)(6x + 7)$
- 41 Find an equation of the tangent line to the graph of $y = 5/(1 + x^2)$ at each point.
(a) $P(0, 5)$ (b) $P(1, \frac{5}{2})$ (c) $P(-2, 1)$
- 42 Find an equation of the tangent line to the graph of $y = 2x^3 + 4x^2 - 5x - 3$ at each point.
(a) $P(0, -3)$ (b) $P(-1, 4)$ (c) $P(1, -2)$
- 43 Find the x -coordinates of all points on the graph of $y = x^3 + 2x^2 - 4x + 5$ at which the tangent line is (a) horizontal; (b) parallel to the line $2y + 8x - 5 = 0$.
- 44 Find point P on the graph of $y = x^3$ such that the tangent line at P has x -intercept 4.

In Exercises 45 and 46 the position of a point P moving on a coordinate line l is given by $f(t)$. Determine the time intervals in which P moves in (a) the positive direction; (b) the negative direction. When is the velocity 0?

45 $f(t) = 2t^3 + 9t^2 - 60t + 1$

46 $f(t) = 3t^5 - 5t^3$

If f and g are differentiable functions such that $f(2) = 3$, $f'(2) = -1$, $g(2) = -5$, and $g'(2) = 2$, find the numbers in Exercises 47 and 48.

- 47 (a) $(f + g)'(2)$ (b) $(f - g)'(2)$
(c) $(4f)'(2)$ (d) $(fg)'(2)$
(e) $(f/g)'(2)$
- 48 (a) $(g - f)'(2)$ (b) $(g/f)'(2)$
(c) $(4g)'(2)$ (d) $(ff)'(2)$

In Exercises 49 and 50 find the points at which the graphs of f and f' intersect.

49 $f(x) = x^3 - x^2 + x + 1$ 50 $f(x) = x^2 + 2x + 1$

- 51 Find an equation of the tangent line to the graph of $y = 3x^2 + 4x - 6$ that is parallel to the line $5x - 2y - 1 = 0$.
- 52 Find equations of the tangent lines to the graph of $y = x^3$ that are parallel to the line $16x - 3y + 17 = 0$.
- 53 Find an equation of the line through the point $P(5, 9)$ that is tangent to the graph of $y = x^2$.
- 54 Find equations of the lines through $P(3, 1)$ that are tangent to the graph of $xy = 4$.
- 55 A weather balloon is released and rises vertically such that its distance $s(t)$ above the ground during the first 10 seconds of flight is given by $s(t) = 6 + 2t + t^2$, where $s(t)$ is in feet and t is in seconds.
(a) Find the velocity of the balloon at $t = 1$, $t = 4$ and $t = 8$.
(b) Find the velocity of the balloon at the instant that it is 50 feet above the ground.
- 56 A ball rolls down an inclined plane such that the distance (in cm) it rolls in t seconds is given by $s(t) = 2t^3 + 3t^2 + 4$, where $0 \leq t \leq 3$ (see figure).
(a) What is the velocity of the ball at $t = 2$?
(b) At what time is the velocity 30 cm/sec?

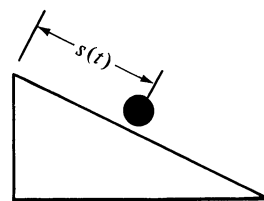


FIGURE FOR EXERCISE 56

3.4 Derivatives of the Sine and Cosine Functions

We shall begin our discussion of derivatives of trigonometric functions by concentrating on sines and cosines. The remaining four trigonometric functions are considered in Chapter 8.

The reader should recall that the graphs of the sine and cosine functions have the shapes illustrated in Figure 3.9. After additional tools of calculus are developed it will be possible to make a detailed analysis of these graphs.

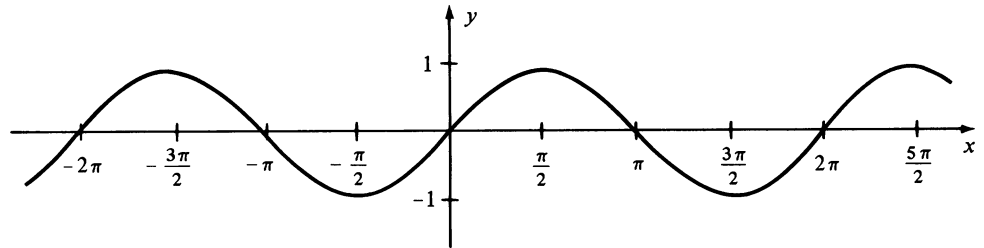
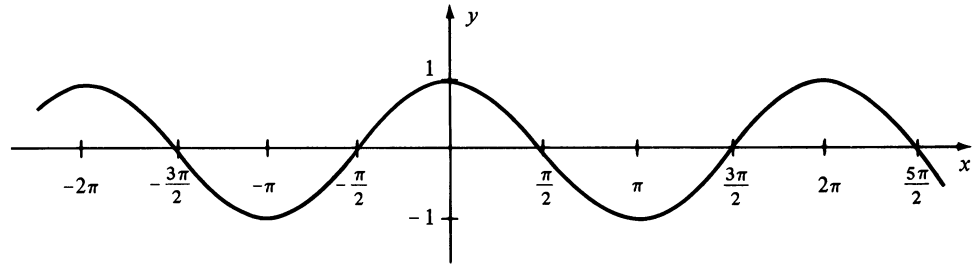

 (i) $y = \sin x$

 (ii) $y = \cos x$

FIGURE 3.9

The formulas for the derivatives of the sine and cosine functions are stated in the next theorem.

Theorem (3.20)

$$D_x \sin x = \cos x$$

$$D_x \cos x = -\sin x$$

Proof To find the derivative of the sine function, we let $f(x) = \sin x$. If $f'(x)$ exists, then

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \end{aligned}$$

Using the addition formula for the sine function (see (1.33)), we obtain

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin x (\cos h - 1) + \cos x \sin h}{h} \\ &= \lim_{h \rightarrow 0} \left[\sin x \left(\frac{\cos h - 1}{h} \right) + \cos x \left(\frac{\sin h}{h} \right) \right]. \end{aligned}$$

By Theorems (2.22) and (2.21),

$$\lim_{h \rightarrow 0} \left(\frac{\cos h - 1}{h} \right) = 0 \quad \text{and} \quad \lim_{h \rightarrow 0} \frac{\sin h}{h} = 1.$$

Hence,

$$f'(x) = (\sin x)(0) + (\cos x)(1) = \cos x,$$

that is,

$$D_x \sin x = \cos x.$$

We have shown that the derivative of the sine function is the cosine function.

In like manner, to obtain the derivative of the cosine function, we let $f(x) = \cos x$. In this case

$$f'(x) = \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}.$$

Applying the addition formula for the cosine function we obtain

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cos x(\cos h - 1) - \sin x \sin h}{h} \\ &= \lim_{h \rightarrow 0} \left[\cos x \left(\frac{\cos h - 1}{h} \right) - \sin x \left(\frac{\sin h}{h} \right) \right]. \end{aligned}$$

Again using Theorems (2.22) and (2.21) we see that

$$f'(x) = (\cos x)(0) - (\sin x)(1) = -\sin x,$$

that is,

$$D_x \cos x = -\sin x. \quad \square$$

Corollary (3.21)

The sine and cosine functions are continuous at every real number.

Proof Since the sine and cosine functions are differentiable, the Corollary is an immediate consequence of Theorem (3.8). $\square$

It follows from the preceding corollary and the Intermediate Value Theorem (2.29), that the sine and cosine functions take on every value between -1 and 1 .

Example 1

- (a) Find the slopes of the tangent lines to the graph of $y = \sin x$ at the points with x -coordinates $0, \pi/3, \pi/2, 2\pi/3$, and π .
- (b) Sketch the tangent lines of part (a) on the graph of $y = \sin x$.

Solution

- (a) The slope of the tangent line at the point (x, y) on the graph of $y = \sin x$ is given by the derivative $y' = \cos x$. In particular, the slopes at the desired points are listed in the following table.

x	0	$\pi/3$	$\pi/2$	$2\pi/3$	π
$y' = \cos x$	1	$1/2$	0	$-1/2$	-1

- (b) The graph of $y = \sin x$ and the tangent lines of part (a) are sketched in Figure 3.10.

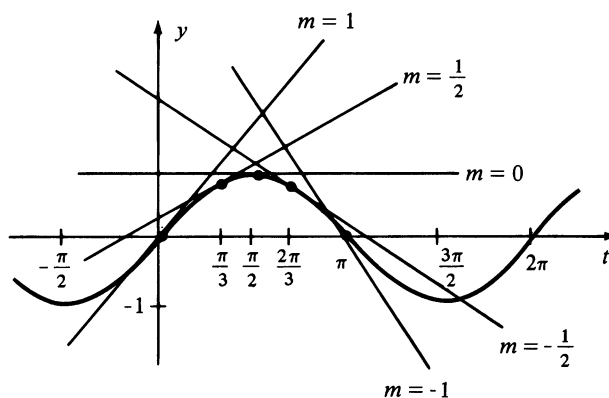


FIGURE 3.10

Example 2 Find y' if $y = \frac{\sin x}{1 + \cos x}$.

Solution By the Quotient Rule and Theorem (3.20),

$$\begin{aligned}
 y' &= \frac{(1 + \cos x)D_x(\sin x) - (\sin x)D_x(1 + \cos x)}{(1 + \cos x)^2} \\
 &= \frac{(1 + \cos x)(\cos x) - \sin x(0 - \sin x)}{(1 + \cos x)^2} \\
 &= \frac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)} \\
 &= \frac{\cos x + 1}{(1 + \cos x)^2} \\
 &= \frac{1}{1 + \cos x}
 \end{aligned}$$

Example 3 Prove that $D_x \tan x = \sec^2 x$.

Solution By a fundamental identity, $\tan x = \sin x / \cos x$. Hence, by the Quotient Rule,

$$\begin{aligned} D_x \tan x &= D_x \left(\frac{\sin x}{\cos x} \right) \\ &= \frac{\cos x (D_x \sin x) - \sin x (D_x \cos x)}{\cos^2 x} \\ &= \frac{\cos x (\cos x) - \sin x (-\sin x)}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} \\ &= \frac{1}{\cos^2 x} = \sec^2 x. \end{aligned}$$

3.4 Exercises

Differentiate the functions defined in Exercises 1–14.

- 1 $f(x) = 4 \cos x$
- 2 $f(x) = 3x \sin x$
- 3 $g(t) = t^3 \sin t$
- 4 $k(t) = t - t^2 \cos t$
- 5 $f(\theta) = (\sin \theta)/\theta$
- 6 $T(r) = r^2/(\cos r)$
- 7 $f(x) = 2x \cos x + x^2 \sin x$
- 8 $f(x) = 3x^2 \sin x - x^3 \cos x$
- 9 $h(z) = (1 - \cos z)/(1 + \cos z)$
- 10 $R(w) = (\cos w)/(1 - \sin w)$
- 11 $g(x) = (x + \sin x)(1 - \cos x)$
- 12 $K(\theta) = (\sin^2 \theta + \cos^2 \theta)^3$
- 13 $p(x) = \sin x \cot x$
- 14 $g(t) = (\sin t - \cos t)^2$
- 15 If $y = x + 2 \cos x$, find
 - (a) the x -coordinates of all points on the graph at which the tangent line is horizontal.
 - (b) the equation of the tangent line to the graph at the point where the graph crosses the y -axis.

- 16 If $y = 3 + 2 \sin x$, find
 - (a) the x -coordinates of all points on the graph where the tangent line is parallel to the line $y = \sqrt{2}x - 5$.
 - (b) an equation of the tangent line to the graph at the point on the graph with x -coordinate $\pi/6$.
- 17 Use (3.20) and the Quotient Rule to prove the following.
 - (a) $D_x \cot x = -\csc^2 x$
 - (b) $D_x \sec x = \sec x \tan x$
 - (c) $D_x \csc x = -\csc x \cot x$
- 18 Prove each of the following.
 - (a) $D_x \sin 2x = 2 \cos 2x$ (*Hint*: $\sin 2x = 2 \sin x \cos x$).
 - (b) $D_x \cos 2x = -2 \sin 2x$ (*Hint*: $\cos 2x = 1 - 2 \sin^2 x$).
- 19 (a) Use the Product Rule to prove that $D_x \sin^2 x = 2 \sin x \cos x$ and $D_x \sin^3 x = 3 \sin^2 x \cos x$.
 (b) Use mathematical induction (see Appendix I) to prove that if n is any positive integer, then

$$D_x \sin^n x = n \sin^{n-1} x \cos x.$$
- 20 Establish formulas that are analogous to those in Exercise 19 for $D_x \cos^2 x$, $D_x \cos^3 x$, and $D_x \cos^n x$.

3.5 Increments and Differentials

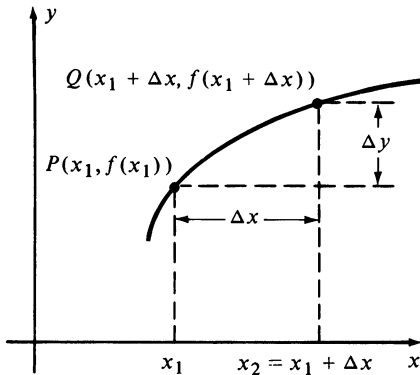


FIGURE 3.11

Let us consider the equation $y = f(x)$ where f is a function. In many applications the independent variable x is subjected to a small change and it is necessary to find the corresponding change in the dependent variable y . If x changes from x_1 to x_2 , then the amount of change is often denoted by the symbol Δx (read “delta x ”); that is,

$$\Delta x = x_2 - x_1.$$

The number Δx is called an **increment** of x . Note that $x_2 = x_1 + \Delta x$; that is, the new value of x equals the initial value plus the change, Δx , in x . Similarly, Δy will denote the change in the dependent variable y that corresponds to the change Δx . Thus

$$\Delta y = f(x_2) - f(x_1) = f(x_1 + \Delta x) - f(x_1).$$

The geometric representations for these increments in terms of the graph of f are shown in Figure 3.11.

Sometimes it is convenient to let x represent the initial value of the independent variable. In this case we use the phrase x is given an increment Δx to describe a (small) change in this variable. Our definition of Δy then has the following form.

Definition (3.22)

Let $y = f(x)$, where f is a function. If x is given an increment Δx , then the **increment Δy** of y is

$$\Delta y = f(x + \Delta x) - f(x).$$

Example 1 Suppose $y = 3x^2 - 5$.

- If x is given an increment Δx , find Δy .
- Use Δy to calculate the numerical change in y if x changes from 2 to 2.1.

Solution

- Let $f(x) = 3x^2 - 5$. Applying Definition (3.22) gives us

$$\begin{aligned} y &= f(x + \Delta x) - f(x) \\ &= [3(x + \Delta x)^2 - 5] - [3x^2 - 5] \\ &= 3[x^2 + 2x(\Delta x) + (\Delta x)^2] - 5 - [3x^2 - 5] \\ &= 3x^2 + 6x(\Delta x) + 3(\Delta x)^2 - 5 - 3x^2 + 5 \\ &= 6x(\Delta x) + 3(\Delta x)^2 \end{aligned}$$

- We wish to find Δy if $x = 2$ and $\Delta x = 0.1$. Substituting in the formula for Δy gives us

$$\begin{aligned} \Delta y &= 6(2)(0.1) + 3(0.1)^2 \\ &= 12(0.1) + 3(0.01) = 1.2 + 0.03 = 1.23. \end{aligned}$$

Thus, y changes by an amount 1.23 if x changes from 2 to 2.01. We could also find Δy directly as follows;

$$\begin{aligned}\Delta y &= f(x + \Delta x) - f(x) = f(2.1) - f(2) \\ &= [3(2.1)^2 - 5] - [3(2)^2 - 5] = 1.23.\end{aligned}$$

■

The increment notation may be used in the definition of the derivative of a function. All that is necessary is to substitute Δx for h in (3.6) as follows:

$$(3.23) \quad f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}.$$

In words (3.23) may be phrased as follows: “The derivative of f is the limit of the ratio of the increment Δy of the dependent variable to the increment Δx of the independent variable as Δx approaches zero.” In Figure 3.11, note that $\Delta y/\Delta x$ is the slope of the secant line through P and Q . It follows from (3.23) that if $f'(x)$ exists, then

$$\frac{\Delta y}{\Delta x} \approx f'(x) \quad \text{if } \Delta x \approx 0.$$

Geometrically this implies that if Δx is close to 0, then the slope $\Delta y/\Delta x$ of the secant line through P and Q is close to the slope $f'(x)$ of the tangent line at P . We may also write

$$\Delta y \approx f'(x) \Delta x \quad \text{if } \Delta x \approx 0.$$

We give the expression $f'(x) \Delta x$ a special name in the next definition.

Definition (3.24)

Let $y = f(x)$, where f is differentiable, and let Δx be an increment of x .

- (i) The **differential dx** of the independent variable x is $dx = \Delta x$.
- (ii) the **differential dy** of the dependent variable y is

$$dy = f'(x) \Delta x = f'(x) dx.$$

Observe that dy depends, for its value, on *both* x and Δx . From (i) we see that as far as the independent variable x is concerned, there is no difference between the increment Δx and the differential dx .

It follows from our previous discussion and (ii) of Definition (3.24) that if Δx is small, then

$$(3.25) \quad \Delta y \approx dy = f'(x) dx = (D_x y) dx.$$

Consequently, if $y = f(x)$, then dy can be used as an approximation to the exact change Δy of the dependent variable corresponding to a small change Δx in x . This observation is useful in applications where only a rough estimate of the change in y is desired.

Example 2 If $y = 3x^2 - 5$, use dy to approximate Δy if x changes from 2 to 2.1.

Solution If $f(x) = 3x^2 - 5$, then by Example 1, $\Delta y = 1.23$. Using Definition (3.24),

$$dy = f'(x) dx = 6x dx.$$

In the present example, $x = 2$, $\Delta x = dx = 0.1$, and

$$dy = (6)(2)(0.1) = 1.2.$$

Hence our approximation is correct to the nearest tenth. ■

Example 3 If $y = x^3$ and Δx is an increment of x , find (a) Δy ; (b) dy ; (c) $\Delta y - dy$; (d) the value of $\Delta y - dy$ if $x = 1$ and $\Delta x = 0.02$.

Solution

(a) Using (3.22) with $f(x) = x^3$ we obtain

$$\Delta y = f(x + \Delta x) - f(x) = (x + \Delta x)^3 - x^3.$$

Applying the Binomial Theorem (3.12) with $n = 3$ gives us

$$\begin{aligned} \Delta y &= x^3 + 3x^2(\Delta x) + 3x(\Delta x)^2 + (\Delta x)^3 - x^3 \\ &= 3x^2(\Delta x) + 3x(\Delta x)^2 + (\Delta x)^3. \end{aligned}$$

(b) By (ii) of Definition (3.24),

$$dy = f'(x) dx = 3x^2 dx = 3x^2(\Delta x).$$

(c) From parts (a) and (b),

$$\begin{aligned} \Delta y - dy &= 3x^2(\Delta x) + 3x(\Delta x)^2 + (\Delta x)^3 - 3x^2(\Delta x) \\ &= 3x(\Delta x)^2 + (\Delta x)^3. \end{aligned}$$

(d) Substituting $x = 1$ and $\Delta x = 0.02$ in part (c),

$$\begin{aligned} \Delta y - dy &= 3(1)(0.02)^2 + (0.02)^3 \\ &= 3(0.0004) + (0.000008) = 0.001208. \end{aligned}$$

This shows that if dy is used to approximate Δy when x changes from 1 to 1.02, then the error involved is approximately 0.001. ■

Example 4 Use differentials to approximate the change in $\sin \theta$ if θ changes from 60° to 61° .

Solution If $y = \sin \theta = f(\theta)$, then

$$dy = f'(\theta) d\theta = \cos \theta d\theta.$$

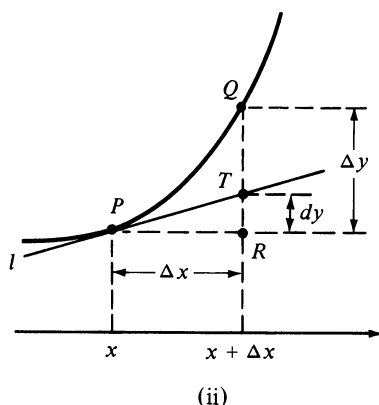
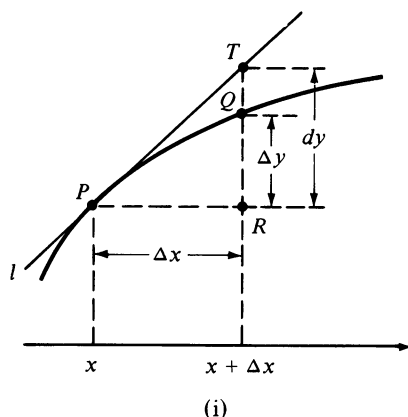


FIGURE 3.12

When we use differentials of trigonometric functions it is essential to employ radian measure. Thus we let $\theta = 60^\circ = \pi/3$ and $\Delta\theta = 1^\circ = \pi/180$ in the formula for dy , obtaining

$$dy = \left(\cos \frac{\pi}{3}\right) \left(\frac{\pi}{180}\right) = \left(\frac{1}{2}\right) \left(\frac{\pi}{180}\right) = \frac{\pi}{360} \approx 0.0087.$$

Using a calculator we find that to four decimal places,

$$\Delta y = \sin 61^\circ - \sin 60^\circ \approx 0.8746 - 0.8660 = 0.0086.$$

Hence the error involved in using dy to approximate Δy is roughly 0.0001. ■

It is instructive to study the geometric interpretations of dx and dy . Consider the graphs in Figure 3.12, where l is the tangent line at the point P . We have pictured both Δx and Δy as positive; however, they may also take on negative values. If T is the point of intersection of l and the vertical line through Q , and if R is the point with coordinates $(x + \Delta x, f(x))$, then it can be seen from triangle PRT that the slope $f'(x)$ of the tangent line is the ratio $r/\Delta x$, where r denotes the length of the segment RT . Hence, $r/\Delta x = f'(x)$ or $r = f'(x) \Delta x = dy$. A similar result holds if increments are negative. In general, if x is given an increment Δx , then dy is the amount that the *tangent line* rises (or falls) when the independent variable changes from x to $x + \Delta x$. This is in contrast to the amount Δy that the *graph* rises (or falls) between P and Q . This geometric interpretation also emphasizes the fact that $dy \approx \Delta y$ if Δx is small.

The next example illustrates how differentials may be used to find certain approximation formulas.

Example 5

- Use differentials to obtain a formula for approximating the volume of a thin cylindrical shell of altitude h , inner radius r , and thickness t .
- What error is involved in using the formula?

Solution

- A typical cylindrical shell is illustrated in Figure 3.13, where the thickness t has been denoted by Δr . We wish to find the volume contained between the “inner” cylinder of radius r and the “outer” cylinder of radius $r + \Delta r$. The volume V of the inner cylinder is given by

$$V = \pi r^2 h.$$

If we increase r by an amount Δr , then the volume of the shell is the change ΔV in V . Using differentials,

$$\Delta V \approx dV = (D_r V) \Delta r = (2\pi r h) \Delta r$$

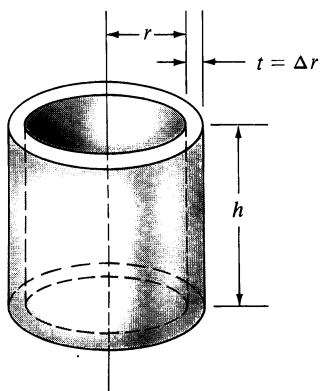


FIGURE 3.13

and hence a formula for approximating the volume of the shell is

$$\Delta V \approx (2\pi rh) \Delta r = (2\pi rh)t.$$

In words,

$$\text{Volume} \approx (\text{Area of inner cylinder wall}) \times (\text{Thickness}).$$

(b) The exact volume of the shell is

$$\Delta V = \pi(r + \Delta r)^2 h - \pi r^2 h$$

which simplifies to

$$\Delta V = (2\pi rh) \Delta r + \pi h(\Delta r)^2.$$

Consequently,

$$\Delta V - dV = \pi h(\Delta r)^2.$$

This indicates that the approximation formula dV is quite accurate if Δr is small in comparison to h . ■

Differentials may sometimes be used to estimate errors, as illustrated by the next example.

Example 6 The radius of a spherical balloon is estimated to be 12 inches with a maximum error in measurement of 0.05 inches. Estimate the maximum error in the calculated volume of the sphere.

Solution If x denotes the radius of the balloon and V the volume, then $V = \frac{4}{3}\pi x^3$. If the maximum error in the measured value of x is denoted by dx (or equivalently, by Δx), then the exact value of the radius is between $x - dx$ and $x + dx$. If we let ΔV denote the error in V caused by the error dx in x , then we may approximate ΔV by means of the differential

$$dV = (D_x V) dx = 4\pi x^2 dx.$$

Letting $x = 12$ and $dx = \pm 0.05$, we obtain

$$dV = 4\pi(12)^2(\pm 0.05) = \pm(28.8)\pi \approx \pm 90.$$

Thus the maximum error in volume due to the error in measurement (± 0.05 inches) of the radius is approximately 90 cubic inches. ■

The terminology introduced in the next definition may be used to describe an error in the measurement of a quantity.

Definition (3.26)

- | |
|---|
| <p>(a) Average error = $\frac{\text{error in measurement}}{\text{measured value}}$</p> <p>(b) Percentage error = (average error) $\times 100$</p> |
|---|

The average error is also called the **relative error**. For example, if the measured length of an object is 20 inches with a possible error of 0.1 inches, then from Definition (3.26), the average error is $0.1/20$ or 0.005 . The significance of this number is that the error involved is, *on the average*, 0.005 inches per inch. The percentage error is defined as the relative error multiplied by 100. In the illustration just given, the percentage error is $(0.005)(100)$ or 0.5% .

In terms of differentials, if w represents a measurement with a possible error of dw , then from our previous remarks, the average error is dw/w . Of course, if dw is an *approximation* to the error in w , then dw/w is an *approximation* to the average error. These remarks are illustrated in the next example.

Example 7 The radius of a spherical balloon is estimated to be 12 inches with a maximum error in measurement of 0.05 inches. Approximate the average error and the percentage error for

- (a) the measured value of the radius.
- (b) the calculated value of the volume.

Solution In Example 6 we estimated the maximum error in the calculated volume. As in that example, we take the measured value of the radius as $x = 12$ inches, with an error in measurement of approximately $dx = \pm 0.05$ inches. Using (3.26) and the error approximation dx gives us the following.

- (a) For the measured value of the radius,

$$\text{Average error} \approx \frac{dx}{x} = \frac{\pm 0.05}{12} \approx \pm 0.004 \text{ inches per inch}$$

$$\text{Percentage error} \approx (\pm 0.004)(100) = \pm 0.4\%$$

- (b) The calculated volume is $V = \frac{4}{3}\pi(12)^3 = 2304\pi$. It was found, in Example 6, that $dV = \pm(28.8)^3\pi$. Hence

$$\text{Average error} \approx \frac{dV}{V} = \frac{\pm(28.8)\pi}{2304\pi} \approx \pm 0.01$$

$$\text{Percentage error} \approx (\pm 0.01)(100) = 1\%. \quad \blacksquare$$

By (ii) of Definition (3.24), $dy = f'(x)dx$. If both sides of this equation are divided by dx (assuming that $dx \neq 0$) we obtain the following notation for the derivative of $y = f(x)$.

$$(3.27) \quad \frac{dy}{dx} = f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

Thus, the derivative $f'(x)$ may be expressed as a quotient of two differentials. Indeed, this is one of the reasons for defining dx and dy as in (3.24). The notation dy/dx was employed by Leibniz in his pioneering work on calculus in the late 1600s and is still widely used in mathematics and the applied

sciences. It is not always essential to regard dy/dx as a quotient of two differentials. As a matter of fact dy/dx is often thought of formally as just another symbol for the derivative $f'(x)$. For example,

$$\text{if } y = 7x^3 - 4x^2 + 2, \text{ then } \frac{dy}{dx} = 21x^2 - 8x.$$

The symbol d/dx is used in the same manner as the operator D_x as follows.

$$(3.28) \quad f'(x) = D_x[f(x)] = \frac{d}{dx}[f(x)] = \frac{d[f(x)]}{dx}$$

In particular, rules for differentiating sums, differences, products, and quotients may be stated as follows, where $u = f(x)$ and $v = g(x)$:

$$(3.29) \quad \begin{aligned} \frac{d}{dx}(u + v) &= \frac{du}{dx} + \frac{dv}{dx} & \frac{d}{dx}(u - v) &= \frac{du}{dx} - \frac{dv}{dx} \\ \frac{d}{dx}(uv) &= u \frac{dv}{dx} + v \frac{du}{dx} & \frac{d}{dx}\left(\frac{u}{v}\right) &= \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} \end{aligned}$$

3.5 Exercises

In Exercises 1–4 (a) use Definition (3.22) to find a general formula for Δy ; (b) use Δy to calculate the change in y corresponding to the stated values of x and Δx .

1 $y = 2x^2 - 4x + 5, x = 2, \Delta x = -0.2$

2 $y = x^3 - 4, x = -1, \Delta x = 0.1$

3 $y = 1/x^2, x = 3, \Delta x = 0.3$

4 $y = 1/(2 + x), x = 0, \Delta x = -0.03$

In Exercises 5–12 find (a) Δy ; (b) dy ; and (c) $dy - \Delta y$.

5 $y = 3x^2 + 5x - 2$

6 $y = 4 - 7x - 2x^2$

7 $y = 1/x$

8 $y = 7x + 12$

9 $y = 4 - 9x$

10 $y = 8$

11 $y = x^4$

12 $y = x^{-2}$

In Exercises 13 and 14 use differentials to approximate the change in $f(x)$ if x changes from a to b .

13 $f(x) = 4x^5 - 6x^4 + 3x^2 - 5; a = 1, b = 1.03$

14 $f(x) = -3x^3 + 8x - 7; a = 4, b = 3.96$

15 If $f(z) = z^3 - 3z^2 + 2z - 7$ and $w = f(z)$, find dw . Use dw to approximate the change in w if z changes from 4 to 3.95.

16 If $F(t) = 1/(2 - t^2)$ and $s = F(t)$, find ds . Use ds to approximate the change in s if t changes from 1 to 1.02.

17 The radius of a circular manhole cover is estimated to be 16 in. with a maximum error in measurement of 0.06 in. Use differentials to estimate the maximum error in the calculated area of one side of the cover. What is the approximate average error and the approximate percentage error?

18 The length of a side of a square floor tile is estimated as 1 ft with a maximum error in measurement of $\frac{1}{16}$ in. Use differentials to estimate the maximum error in the calculated area. What is the approximate average error and the approximate percentage error?

19 Use differentials to approximate the increase in volume of a cube if the length of each edge changes from 10 in. to 10.1 in. What is the exact change in volume?

20 A spherical balloon is being inflated with gas. Use differentials to approximate the increase in surface area of the balloon if the diameter changes from 2 ft to 2.02 ft.

21 One side of a house has the shape of a square surmounted by an equilateral triangle. If the length of the base is measured as 48 ft, with a maximum error in measurement of 1 in., calculate the area of the side and use differentials to estimate the maximum error in the calculation. What is the approximate average error and the approximate percentage error?

- 22 Small errors in measurements of dimensions of large containers can have a marked effect on calculated volumes. A silo has the shape of a right circular cylinder surmounted by a hemisphere (see figure). The altitude of the cylinder is exactly 50 ft. The circumference of the base is measured as 30 ft, with a maximum error in measurement of 6 in. Calculate the volume of the silo from these measurements and use differentials to estimate the maximum error in the calculation. What is the approximate average error and the approximate percentage error?



FIGURE FOR EXERCISE 22

- 23 As sand leaks out of a container, it forms a conical pile whose altitude is always the same as the radius. If, at a certain instant, the radius is 10 cm, use differentials to approximate the change in radius that will increase the volume of the pile by 2 cm^3 .
- 24 Use the technique illustrated in Example 5 of this section to find formulas that can be used to approximate the volume of a thin shell-shaped solid whose surface is (a) spherical; (b) cubical.
- 25 Newton's Law of Gravitation states that the force F of attraction between two particles having masses m_1 and m_2 is given by $F = gm_1m_2/s^2$, where g is a constant and s is the distance between the particles. If $s = 20 \text{ cm}$, use differentials to approximate the change in s that will increase F by 10% .
- 26 Boyle's Law states that the pressure p and volume v of a confined gas are related by the formula $pv = c$, where c is a constant, or equivalently, by $p = c/v$, where $v \neq 0$. Show that dp and dv are related by means of the formula $p dv + v dp = 0$.
- 27 Use differentials to approximate $(0.98)^4$. (Hint: Let $y = f(x) = x^4$ and consider $f(x + \Delta x) = f(x) + \Delta y$, where $x = 1$ and $\Delta x = -0.02$.) What is the exact value of $(0.98)^4$?
- 28 Use differentials to approximate
- $$N = (2.01)^4 - 3(2.01)^3 + 4(2.01)^2 - 5.$$
- What is the exact value of N ?
- 29 The area A of a square of side s is given by $A = s^2$. If s increases by an amount Δs , give geometric illustrations of dA and $\Delta A - dA$.
- 30 The volume V of a cube of edge x is given by $V = x^3$. If x increases by an amount Δx , give geometric illustrations of dV and $\Delta V - dV$.
- 31 Constriction of arterioles is a cause of high blood pressure. It has been verified experimentally that as blood flows through an arteriole of fixed length, the pressure difference between the two ends of the arteriole is inversely proportional to the fourth power of the radius. If the radius of an arteriole decreases by 10% , use differentials to find the percentage change in the pressure difference.
- 32 The electrical resistance R of a wire is directly proportional to its length and inversely proportional to the square of its diameter. If the length is fixed, how accurately must the diameter be measured (in terms of percentage error) to keep the percentage error in R between -3% and 3% ?
- 33 Use differentials to approximate the change in $\cos \theta$ if θ changes from 30° to 28° .
- 34 An isosceles triangle has equal sides of length 12 in. If the angle θ between these sides is increased from 30° to 33° , use differentials to approximate the change in the area of the triangle.
- 35 If an object of weight W lb is pulled along a horizontal plane by a force applied to a rope that is attached to the object, and if the rope makes an angle θ with the horizontal, then it can be shown that the magnitude of the force is given by

$$F = \frac{\mu W}{\mu \sin \theta + \cos \theta}$$

where μ is a constant called the *coefficient of friction*. Suppose a man is pulling a 100-pound box along a floor, and that $\mu = 0.2$ (see figure). If θ is changed from 45° to 46° , use differentials to approximate the change in the force that must be applied.

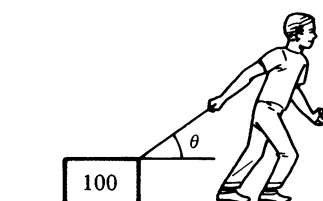


FIGURE FOR EXERCISE 35

- 36 It will be shown in Chapter 15 that if a projectile is fired from a cannon with an initial velocity v_0 and at an angle α to the horizontal, then its maximum height h and range R are given by

$$h = \frac{v_0^2 \sin^2 \alpha}{2g} \quad \text{and} \quad R = \frac{2v_0^2 \sin \alpha \cos \alpha}{g}.$$

Suppose $v_0 = 100$ ft/sec and $g = 32$ ft/sec². If α is increased from 30° to $30^\circ 30''$, use differentials to estimate the changes in h and R .

3.6 The Chain Rule

The rules for derivatives obtained in previous sections are limited in scope because they can only be used for sums, differences, products, and quotients that involve x^n , $\cos x$, or $\sin x$. At this stage of our work there is no rule that can be applied *directly* to an expression such as $\sin 2x$. Clearly,

$$D_x \sin 2x \neq \cos 2x,$$

for if we let

$$y = \sin 2x = 2 \sin x \cos x$$

and apply the Product Rule, we obtain

$$\begin{aligned} D_x y &= 2D_x(\sin x \cos x) \\ &= 2[\sin x(D_x \cos x) + \cos x(D_x \sin x)] \\ &= 2[\sin x(-\sin x) + \cos x(\cos x)] \\ &= 2[-\sin^2 x + \cos^2 x] = 2 \cos 2x. \end{aligned}$$

Hence,

$$D_x \sin 2x = 2 \cos 2x.$$

Since the preceding manipulations are rather cumbersome, let us seek a more direct method of finding the derivative. The key to our search is to express y as a *composite* function of x . Recall that if f and g are functions, where

$$y = f(u) \quad \text{and} \quad u = g(x),$$

and if $g(x)$ is in the domain of f , then we may write

$$y = f(u) = f(g(x)),$$

that is, y is a function of x . Indeed, the function is the same as the composite function $f \circ g$ defined in Section 1.5. Note that $\sin 2x$ may be expressed in this way by letting

$$y = \sin u \quad \text{and} \quad u = 2x.$$

Thus, if we can obtain a general rule for differentiating $f(g(x))$, then it may be applied to $y = \sin 2x$ and, in fact, to any expression of the form $y = \sin g(x)$.

To get some idea of the type of rule to expect, let us return to the equations

$$y = f(u) \quad \text{and} \quad u = g(x).$$

Our goal is to find a formula for the derivative dy/dx of the composite function given by $y = f(g(x))$. If f and g are differentiable, then using the differential notation (3.27),

$$\frac{dy}{du} = f'(u) \quad \text{and} \quad \frac{du}{dx} = g'(x).$$

If we consider the product

$$\frac{dy}{du} \frac{du}{dx}$$

and treat the derivatives as quotients of differentials, then the product suggests the following rule:

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = f'(u)g'(x).$$

Note that this rule does, in fact, lead to the correct derivative of $y = \sin 2x$, for if we write

$$y = \sin u \quad \text{and} \quad u = 2x$$

and use the rule, we obtain

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = (\cos u)(2) = 2 \cos u = 2 \cos 2x.$$

Although we have not *proved* that this technique is valid, it makes the next theorem seem extremely plausible. In the statement of the theorem it is assumed that variables are chosen such that the composite function $f \circ g$ is defined, and that if g has a derivative at x , then f has a derivative at $g(x)$.

The Chain Rule (3.30)

If $y = f(u)$, $u = g(x)$, and the derivatives $\frac{dy}{du}$ and $\frac{du}{dx}$ both exist, then the composite function defined by $y = f(g(x))$ has a derivative given by

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = f'(u)g'(x) = f'(g(x))g'(x).$$

Partial Proof Let Δx be an increment such that both x and $x + \Delta x$ are in the domain of the composite function. Since $y = f(g(x))$, the corresponding increment of y is given by

$$\Delta y = f(g(x + \Delta x)) - f(g(x)).$$

If the composite function has a derivative at x , then by (3.27),

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}.$$

Next consider $u = g(x)$ and let Δu be the increment of u that corresponds to Δx , that is,

$$\Delta u = g(x + \Delta x) - g(x).$$

Since $g(x + \Delta x) = g(x) + \Delta u = u + \Delta u$

we may express the formula for Δy as

$$\Delta y = f(u + \Delta u) - f(u).$$

If $y = f(u)$ is differentiable at u , then

$$\frac{dy}{du} = f'(u) = \lim_{\Delta u \rightarrow 0} \frac{\Delta y}{\Delta u}.$$

Similarly, if $u = g(x)$ is differentiable at x , then

$$\frac{du}{dx} = g'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x}.$$

Let us assume that there exists an open interval I containing x such that whenever $x + \Delta x$ is in I and $\Delta x \neq 0$, then $\Delta u \neq 0$. In this case we may write

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta y}{\Delta u} \frac{\Delta u}{\Delta x} \right) = \left(\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta u} \right) \left(\lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} \right)$$

provided the limits exist. Since g is differentiable at x , it is continuous at x . Hence if $\Delta x \rightarrow 0$, then $g(x + \Delta x)$ approaches $g(x)$ and, therefore, $\Delta u \rightarrow 0$. It follows that the last limit formula displayed above may be written

$$\begin{aligned} \frac{dy}{dx} &= \left(\lim_{\Delta u \rightarrow 0} \frac{\Delta y}{\Delta u} \right) \left(\lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} \right) \\ &= \left(\frac{dy}{du} \right) \left(\frac{du}{dx} \right) = f'(u)g'(x) = f'(g(x))g'(x) \end{aligned}$$

which is what we wished to prove.

In most applications of the Chain Rule, $u = g(x)$ has the property assumed in the preceding paragraph. If g does not satisfy this property, then every open interval containing x contains a number $x + \Delta x$, with $\Delta x \neq 0$, such that $\Delta u = 0$. In this case our proof is invalid, since Δu occurs in a denominator. To construct a proof that takes functions of this type into account, it is necessary to introduce techniques which are more sophisticated than those we have used. A complete proof of the Chain Rule is given in Appendix II. $\square$

Example 1 If $y = u^3$ and $u = x^2 + 1$, find dy/dx .

Solution The problem could be worked by substituting $x^2 + 1$ for u in the first equation, obtaining $y = (x^2 + 1)^3 = x^6 + 3x^4 + 3x^2 + 1$, and then calculating dy/dx . Another approach is to use the Chain Rule (3.30). Thus

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = (3u^2)(2x).$$

If $x^2 + 1$ is substituted for u we obtain

$$\frac{dy}{dx} = 3(x^2 + 1)^2(2x). \quad \blacksquare$$

One of the main uses for the Chain Rule is to establish other differentiation formulas. As a first illustration we shall obtain a formula for the derivative of a power of a function.

The Power Rule for Functions (3.31)

If g is a differentiable function and n is an integer, then

$$D_x[g(x)]^n = n[g(x)]^{n-1}D_x[g(x)].$$

Proof If we let $y = u^n$ and $u = g(x)$, then $y = [g(x)]^n$ and, by the Chain Rule (3.30),

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = nu^{n-1}D_x u = n[g(x)]^{n-1}D_x[g(x)].$$

This completes the proof. □

The following form of the Power Rule for Functions is easier to remember than (3.31):

$$(3.32) \quad D_x(u^n) = nu^{n-1}D_x u$$

where $u = g(x)$ and n is an integer. Note that if $u = x$, then $D_x u = 1$ and (3.32) reduces to (3.13).

Example 2 Find $f'(x)$ if $f(x) = (x^5 - 4x + 8)^7$.

Solution Using (3.32) with $u = x^5 - 4x + 8$ and $n = 7$,

$$\begin{aligned} f'(x) &= D_x(x^5 - 4x + 8)^7 = 7(x^5 - 4x + 8)^6 D_x(x^5 - 4x + 8) \\ &= 7(x^5 - 4x + 8)^6(5x^4 - 4). \end{aligned} \quad \blacksquare$$

Example 3 Find y' if $y = \frac{1}{(4x^2 + 6x - 7)^3}$.

Solution Writing $y = (4x^2 + 6x - 7)^{-3}$ and applying the Power Rule (3.32) with $u = 4x^2 + 6x - 7$ and $n = -3$,

$$\begin{aligned} y' &= D_x(4x^2 + 6x - 7)^{-3} \\ &= -3(4x^2 + 6x - 7)^{-4} D_x(4x^2 + 6x - 7) \\ &= -3(4x^2 + 6x - 7)^{-4} (8x + 6) \\ &= \frac{-6(4x + 3)}{(4x^2 + 6x - 7)^4}. \end{aligned}$$

Example 4 If $F(z) = (2z + 5)^3(3z - 1)^4$, find $F'(z)$.

Solution Using first the Product Rule, second the Power Rule, and then factoring the result, we have

$$\begin{aligned} F'(z) &= (2z + 5)^3 D_z(3z - 1)^4 + (3z - 1)^4 D_z(2z + 5)^3 \\ &= (2z + 5)^3 \cdot 4(3z - 1)^3(3) + (3z - 1)^4 \cdot 3(2z + 5)^2(2) \\ &= 6(2z + 5)^2(3z - 1)^3[2(2z + 5) + (3z - 1)] \\ &= 6(2z + 5)^2(3z - 1)^3(7z + 9). \end{aligned}$$

The Chain Rule will have far-reaching consequences later in the text, in our work with differentiation and integration. Two important formulas which may be obtained now are stated in the next theorem.

Theorem (3.33)

If $u = g(x)$ where g is a differentiable function, then

$$\begin{aligned} D_x \sin u &= (\cos u) D_x u \\ D_x \cos u &= (-\sin u) D_x u \end{aligned}$$

Proof If we let $y = \sin u$, then by (3.20),

$$\frac{dy}{du} = \frac{d}{du} (\sin u) = \cos u.$$

Applying the Chain Rule (3.30),

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = (\cos u) D_x u.$$

The formula for $D_x \cos u$ may be obtained in similar fashion. □

Example 5 Find y' if $y = \cos(5x^3)$.

Solution Using the formula for $D_x \cos u$ in Theorem (3.33) with $u = 5x^3$ gives us

$$\begin{aligned} D_x \cos(5x^3) &= [-\sin(5x^3)]D_x(5x^3) \\ &= [-\sin(5x^3)](15x^2) \\ &= -15x^2 \sin(5x^3). \end{aligned}$$

Example 6 Find $f'(x)$ if $f(x) = \sin^3(4x)$.

Solution First note that $f(x) = \sin^3(4x) = [\sin(4x)]^3$. Applying the Power Rule for Functions (3.32) with $u = \sin(4x)$ we obtain

$$f'(x) = 3[\sin(4x)]^2 D_x(\sin 4x) = (3 \sin^2 4x) D_x \sin 4x.$$

Next, by Theorem (3.33),

$$D_x \sin 4x = (\cos 4x) D_x(4x) = (\cos 4x)(4) = 4 \cos 4x.$$

Consequently,

$$f'(x) = (3 \sin^2 4x)(4 \cos 4x) = 12 \sin^2 4x \cos 4x.$$

If g is a differentiable function and $y = g(x)$, then (3.32) may be written in any of the forms

$$(3.34) \quad D_x(y^n) = ny^{n-1} D_x y = ny^{n-1} y' = ny^{n-1} \frac{dy}{dx}.$$

The dependent variable y represents the expression $g(x)$ and consequently, when differentiating y^n it is *essential* to multiply ny^{n-1} by the derivative y' . A common error is to write $D_x y^n = ny^{n-1}$. Formula (3.34) will be extremely important in our work with implicit functions in the next section.

3.6 Exercises

Differentiate the functions defined in Exercises 1–38.

1 $f(x) = (x^2 - 3x + 8)^3$

2 $f(x) = (4x^3 + 2x^2 - x - 3)^2$

3 $g(x) = (8x - 7)^{-5}$

4 $k(x) = (5x^2 - 2x + 1)^{-3}$

5 $f(x) = x/(x^2 - 1)^4$

6 $g(x) = (x^4 - 3x^2 + 1)/(2x + 3)^4$

7 $f(x) = (8x^3 - 2x^2 + x - 7)^5$

8 $g(w) = (w^4 - 8w^2 + 15)^4$

9 $F(v) = (17v - 5)^{1000}$

10 $K(x) = (3x^2 - 5x + 7)^{-1}$

11 $s(t) = (4t^5 - 3t^3 + 2t)^{-2}$

12 $p(s) = 1/(8 - 5s + 7s^2)^{10}$

13 $k(x) = \sin(5x + 2)$

14 $f(t) = \cos(4 - 3t)$

15 $g(x) = \cos(2x^2 - 3x + 1)$

16 $G(x) = \sin(2x^5)$

17 $f(\theta) = \cos^4 3\theta$

18 $h(u) = \sin^3(u^4)$

19 $r(t) = \sin^2 t + \sin t^2$

20 $f(x) = (\sin 3x)(\cos 4x)$

21 $f(x) = x^3 \cos(1/x)$

22 $T(\theta) = (\theta^4 + \cos^4 \theta)^4$

23 $G(v) = \frac{\sin 3v}{\sin 3v + \cos 3v}$

24 $F(t) = \frac{\sin(3t + 4)}{3t + 4}$

25 $N(x) = (6x - 7)^3(8x^2 + 9)^2$

26 $f(w) = (2w^2 - 3w + 1)(3w + 2)^4$

27 $g(z) = \left(z^2 - \frac{1}{z^2}\right)^6$

28 $S(t) = \left(\frac{3t + 4}{6t - 7}\right)^3$

29 $k(u) = (u^2 + 1)^3/(4u - 5)^5$

30 $g(x) = (3x - 8)^{-2}(7x^2 + 4)^{-3}$

31 $f(x) = \left(\frac{3x^2 - 5}{2x^2 + 7}\right)^2$

32 $M(z) = \frac{9z^3 + 2z}{(6z + 1)^3}$

33 $G(s) = (s^{-4} + 3s^{-2} + 2)^{-6}$

34 $F(v) = (v^{-1} - 2v^{-2})^{-3}$

35 $h(x) = [(2x + 1)^{10} + 1]^{10}$

36 $f(t) = \left[\left(1 + \frac{1}{t}\right)^{-1} + 1\right]^{-1}$

37 $F(t) = 2t(2t + 1)^2(2t + 3)^3$

38 $N(x) = \frac{7x(x^2 + 1)^2}{(3x + 10)^4}$

For Exercises 39–42 find (a) an equation of the tangent line to the graph of the given equation at the indicated point; and (b) the points on the graph at which the tangent line is horizontal.

39 $y = (4x^2 - 8x + 3)^4, P(2, 81)$

40 $y = \left(x + \frac{1}{x}\right)^5, P(1, 32)$

41 $y = (2x - 1)^{10}, P(1, 1)$

42 $y = (x^2 - 1)^7, P(0, -1)$

43 Find an equation of the tangent line to the graph of $y = x^2 \sin 4x$ at the point with x -coordinate $\pi/4$.

44 Find the x -coordinates of all points on the graph of $y = \sin 3x + \cos 3x$ at which the tangent line is horizontal.

45 A cork bobs up and down in a pail of water such that the distance from the bottom of the pail to the center of the cork at time $t \geq 0$ is given by $s(t) = 12 + \cos(\pi t)$, where $s(t)$ is in inches and t is in seconds (see figure).

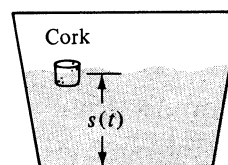


FIGURE FOR EXERCISE 45

(a) Find the velocity of the cork at $t = 0$, $t = \frac{1}{2}$, $t = 1$, $t = \frac{3}{2}$ and $t = 2$.

(b) During what time intervals is the cork rising? When is it falling?

46 A particle in a vibrating spring is moving vertically such that its distance $s(t)$ from a fixed point on the line of vibration is given by $s(t) = 4 + \frac{1}{25} \sin(100\pi t)$, where $s(t)$ is in cm and t is in seconds.

(a) How long does it take the particle to make one complete vibration?

(b) Find the velocity of the particle at $t = 1$, $t = 1.005$, $t = 1.01$, and $t = 1.015$.

47 If $y = (x^4 - 3x^2 + 1)^{10}$, find dy and use it to approximate the change in y if x changes from 1 to 1.01.

48 If $w = z^3(z - 1)^5$, find dw and use it to approximate the change in w if z changes from 2 to 1.98.

49 If $y = \cos^3 2\theta$, find dy and use it to approximate the change in y if θ changes from 30° to $28^\circ 30'$.

50 Use the Chain Rule, the derivative formula for $D_x \sin u$, together with the identities

$$\cos x = \sin\left(\frac{\pi}{2} - x\right) \quad \text{and} \quad \sin x = \cos\left(\frac{\pi}{2} - x\right)$$

to obtain the formula for $D_x \cos x$.

51 If $w = f(z)$ and $z = g(s)$, (a) express the Chain Rule formula for dw/ds in terms of the differential notation; and (b) find dw/ds if $w = z^3 - (2/z)$ and $z = (s^2 + 1)^5$.

- 52 If $v = F(u)$ and $u = G(t)$, (a) express the Chain Rule formula for dv/dt in terms of the differential notation; and (b) find dv/dt if $v = (u^4 + 2u^2 + 1)^3$ and $u = 4t^2$.
- 53 If a body of mass m moves rectilinearly with a velocity v , then its *kinetic energy* K is given by $K = \frac{1}{2}mv^2$. If v is a function of time t , use the Chain Rule to find a formula for dK/dt .
- 54 As a spherical weather balloon is being inflated, its radius r is a function of time t . If V is the volume of the balloon, use the Chain Rule to find a formula for dV/dt .
- 55 If $f(x) = x^4 - 3x^3 + 3x + 2$ and $g(x) = x^3 - 3x^2 + 2x + 1$, use differentials to approximate the change in $f(g(x))$ if x changes from 1 to 0.99.
- 56 If $f(x) = x^5 + x^4 + x^3 + x^2 + x + 1$, approximate the change in $f(f(x))$ if x changes from 0 to 0.01.
- 57 If $k(x) = f(g(x))$ with $f(2) = -4$, $g(2) = 2$, $f'(2) = 3$, and $g'(2) = 5$, find $k'(2)$.
- 58 If r , s , and t are functions such that $r(x) = s(t(x))$, and if $s(0) = -1$, $t(0) = 0$, $s'(0) = -3$, and $r'(0) = 2$, find $t'(0)$.
- 59 If $z = k(y)$, $y = f(u)$, and $u = g(x)$, show that under suitable restrictions

$$\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{du} \frac{du}{dx}.$$

Extend this result to any number of functions.

- 60 A function f is *even* if $f(-x) = f(x)$ for all x in its domain D , whereas f is *odd* if $f(-x) = -f(x)$ for all x in D . Suppose f is differentiable. Use the Chain Rule to prove that (a) if f is even, then f' is odd, and, (b) if f is odd, then

f' is even. Use polynomial functions and the sine and cosine functions to give examples of (a) and (b).

- 61 The **normal line** at a point $P(x_1, y_1)$ on the graph of a differentiable function f is defined as the line through P that is perpendicular to the tangent line. If $f'(x_1) \neq 0$, prove that an equation of the normal line at $P(x_1, y_1)$ is

$$y - y_1 = -\frac{1}{f'(x_1)}(x - x_1).$$

- 62 Refer to Exercise 61. What is an equation of the normal line if $f'(x_1) = 0$?

In Exercises 63–66 find an equation of the normal line to the graph of f at the indicated point P . Sketch the graph, showing the normal line at P .

- 63 $f(x) = x^2 + 1$, $P(1, 2)$
- 64 $f(x) = 8 - x^3$, $P(1, 7)$
- 65 $f(x) = 1/(x^2 + 1)$, $P(1, \frac{1}{2})$
- 66 $f(x) = (x - 1)^4$, $P(2, 1)$
- 67 Find an equation of the line with slope 4 that is normal to the graph of $y = 1/(8x + 3)^2$.
- 68 Find the points on the graph of $y = (2x - 1)^3$ at which the normal line is parallel to the line $x + 12y = 36$.
- 69 In Chapter 7 a function denoted by $\ln$ is defined which has the property $D_x \ln x = 1/x$ for all $x > 0$. Use the Chain Rule to find $D_x \ln(x^4 + 2x^2 + 5)$.
- 70 In Chapter 7 a function f is defined which has the property $D_x[f(x)] = f(x)$ for all x . Find a formula for $D_x[f(g(x))]$, where g is any differentiable function.

3.7 Implicit Differentiation

Given an equation of the form

$$y = 2x^2 - 3$$

it is customary to say that y is a *function of* x , since we may write

$$y = f(x), \quad \text{where } f(x) = 2x^2 - 3.$$

The equation

$$4x^2 - 2y = 6$$

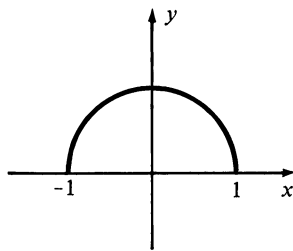
defines the same function f , since solving for y gives us

$$-2y = -4x^2 + 6, \quad \text{or } y = 2x^2 - 3.$$

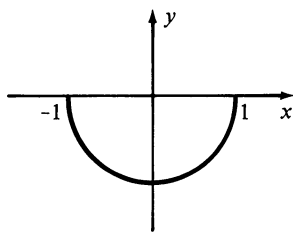
In the case $4x^2 - 2y = 6$ we say that y is an **implicit function** of x , or that f is defined *implicitly* by the equation. Substituting $f(x)$ for y in $4x^2 - 2y = 6$ we obtain

$$\begin{aligned} 4x^2 - 2f(x) &= 6 \\ 4x^2 - 2(2x^2 - 3) &= 6 \\ 4x^2 - 4x^2 + 6 &= 6 \end{aligned}$$

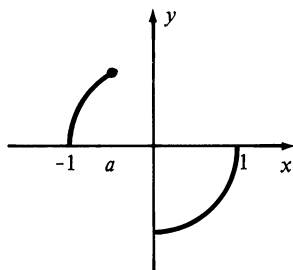
which is an identity since it is true for every x in the domain of f . This fact is characteristic of every function f defined implicitly by an equation in x and y ; that is, f is implicit if and only if substitution of $f(x)$ for y leads to an identity. Since $(x, f(x))$ is a point on the graph of f , the last statement implies that *the graph of the implicit function f coincides with a portion (or all) of the graph of the equation*. This is illustrated in the next example, where it is also shown that an equation in x and y may determine more than one implicit function.



(i)



(ii)



(iii)

FIGURE 3.14

Example 1 How many different functions are determined implicitly by the equation $x^2 + y^2 = 1$?

Solution The graph of $x^2 + y^2 = 1$ is the unit circle with center at the origin. Solving the equation for y in terms of x we obtain

$$y = \pm \sqrt{1 - x^2}.$$

Two obvious implicit functions are given by

$$f(x) = \sqrt{1 - x^2} \quad \text{and} \quad g(x) = -\sqrt{1 - x^2}.$$

The graphs of f and g are the upper and lower halves, respectively, of the unit circle (see (i) and (ii) of Figure 3.14). To find other implicit functions we may let a be any number between -1 and 1 and then define the function k by

$$k(x) = \begin{cases} \sqrt{1 - x^2} & \text{if } -1 \leq x \leq a \\ -\sqrt{1 - x^2} & \text{if } a < x \leq 1. \end{cases}$$

The graph of k is sketched in (iii) of Figure 3.14. Note that there is a discontinuity at $x = a$. The function k is defined implicitly by $x^2 + y^2 = 1$, since

$$x^2 + [k(x)]^2 = 1$$

for every x in the domain of k . By letting a take on different values, it is possible to obtain as many implicit functions as desired. There are also many other types of functions which are determined implicitly by $x^2 + y^2 = 1$. As mentioned previously, the graph of each implicit function is a portion of the graph of the unit circle. ■

It is not obvious that an equation such as

$$y^4 + 3y - 4x^3 = 5x + 1$$

determines an implicit function f in the sense that if $f(x)$ is substituted for y , then

$$[f(x)]^4 + 3[f(x)] - 4x^3 = 5x + 1$$

is an identity for all x in the domain of f . It is possible to state conditions under which an implicit function exists and is differentiable at numbers in its domain; however, the proof requires advanced methods and hence is omitted here. In the examples that follow it will be assumed that a given equation in x and y determines a differentiable function f such that if $f(x)$ is substituted for y , the equation is an identity for all x in the domain of f . The derivative of f may then be found by the method of **implicit differentiation** as illustrated in the following example.

Example 2 Assuming that the equation $y^4 + 3y - 4x^3 = 5x + 1$ determines, implicitly, a differentiable function f such that $y = f(x)$, find its derivative.

Solution We think of y as a symbol which denotes $f(x)$ and regard the equation as an identity for all x in the domain of f . It follows that the derivatives of both sides are equal, that is,

$$\begin{aligned} D_x(y^4 + 3y - 4x^3) &= D_x(5x + 1), \quad \text{or} \\ (*) \quad D_x(y^4) + D_x(3y) - D_x(4x^3) &= D_x(5x) + D_x(1). \end{aligned}$$

It is important to remember that, in general, $D_x y^4 \neq 4y^3$. Instead, since $y = f(x)$, we use (3.34), obtaining $D_x(y^4) = 4y^3 y'$. In like manner, $D_x(3y) = 3D_x y = 3y'$. Substituting in (*),

$$4y^3 y' + 3y' - 12x^2 = 5 + 0.$$

We now solve for y' . Thus

$$(4y^3 + 3)y' = 12x^2 + 5$$

and

$$y' = \frac{12x^2 + 5}{4y^3 + 3}$$

provided the denominator is not zero. In terms of f , this becomes

$$f'(x) = \frac{12x^2 + 5}{4(f(x))^3 + 3}. \quad \blacksquare$$

In the solution of Example 2 the last two equations bring out a disadvantage in using the method of implicit differentiation, namely that the formula for y' (or $f'(x)$) may contain the expression y (or $f(x)$). However, these formulas can still be extremely useful in analyzing f and its graph.

In the next example, implicit differentiation is used to find the slope of the tangent line at a point $P(a, b)$ on the graph of an equation. In problems of this type we shall assume that the equation defines an implicit function f

whose graph coincides with the graph of the equation for all x in some open interval containing a . It is also important to note that, since $P(a, b)$ is a point on the graph, the ordered pair (a, b) must be a solution of the equation.

Example 3 Find the slope of the tangent line to the graph of

$$y^4 + 3y - 4x^3 = 5x + 1$$

at the point $P(1, -2)$.

Solution Note that $P(1, -2)$ is on the graph since

$$(-2)^4 + 3(-2) - 4(1)^3 = 5(1) + 1.$$

By Definition (3.1) the slope m of the tangent line at $P(1, -2)$ is the value of the derivative y' when $x = 1$ and $y = -2$. The given equation is the same as that in Example 2, where it was found that $y' = (12x^2 + 5)/(4y^3 + 3)$. Substituting 1 for x and -2 for y gives us

$$m = \frac{12(1)^2 + 5}{4(-2)^3 + 3} = -\frac{17}{29}. \quad \blacksquare$$

Example 4 Assume that $x^2 + y^2 = 1$ determines a function f such that $y = f(x)$. Find y' .

Solution The given equation was considered in Example 1. Proceeding as in Example 2, we differentiate both sides with respect to x , obtaining

$$D_x(x^2) + D_x(y^2) = D_x(1).$$

Consequently,

$$2x + 2yy' = 0, \quad \text{or} \quad yy' = -x$$

and
$$y' = -\frac{x}{y}, \quad \text{if } y \neq 0. \quad \blacksquare$$

The equation $x^2 + y^2 = 1$ determines many implicit functions (see Example 1). It can be shown that the method of implicit differentiation provides the derivative of *any* differentiable function determined by an equation in two variables. To illustrate, the slope of the tangent line at the point (x, y) on any of the graphs in Figure 3.14 is given by $y' = -x/y$, provided the derivative exists.

Example 5 Find y' if $4xy^3 - x^2y + x^3 - 5x + 6 = 0$.

Solution Differentiating both sides of the equation with respect to x we obtain

$$D_x(4xy^3) - D_x(x^2y) + D_x(x^3) - D_x(5x) + D_x(6) = D_x(0).$$